

BESOV-ISH SPACES THROUGH ATOMIC DECOMPOSITION: THE RESULTS FORMALIZED IN LEAN

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ABSTRACT. This document describes, without proofs, exactly those results of the paper *Besov-ish spaces through atomic decomposition* (Analysis & PDE 15 (2022), no. 1; LaTeX source `besovish-2021-07-19.tex`) that have been formally verified in the Lean 4 repository `BesovSpacesGoodGrid`, together with a few results obtained in the formalization that go beyond the paper. Statements follow the notation of the paper; footnotes record every deviation from the published statements, and each result is annotated with the names of the corresponding Lean declarations. The repository compiles with no `sorry`, and the main theorems depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`.

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1. INTRODUCTION

This is a mathematical description of the current state (June 9, 2026) of the Lean 4 formalization of [1]. It was produced from the LaTeX source of the published paper by the following rules.

- Results that are not present in the repository were deleted. These are: the elementary estimates of paper Section 4 (Hölder-like and convolution tricks, inlined in the Lean proofs without standalone statements); the Hölder and bounded variation atoms (paper Sections 11B–11C) and the corresponding characterizations (paper Propositions 16.2 and 16.3); Dirac’s approximations (paper Section 17); Corollary 18.6, strongly regular domains (paper 18.7–18.8, except for the grid-cell case, which is formalized), Pointwise Multipliers I and II (paper 18.9–18.10); and all of paper Sections 19–21 (quasialgebra, description of $\mathcal{B}_{1,1}^s$ by indicators, left compositions).
- Statements were updated to match the precision of the formalization; every change with mathematical content is recorded in a footnote comparing with `besovish-2021-07-19.tex`.
- No proofs are included. For each statement the corresponding Lean declarations and files are given in a footnote, and the statement is followed by a snapshot of the Lean code that formalizes it (the formal statement as it appears in the repository, with the proof omitted; an ellipsis $\dots$ marks the omitted proof or, in a few long definitions, omitted structural fields).

A global change must be singled out: *throughout this document* $p \in [1, \infty)$, $q \in [1, \infty]$ and $u \in [1, \infty]$. The paper develops Part I for the quasi-Banach range $p, q \in (0, \infty]$; the formalization restricts to the Banach range, so $\rho = \min\{1, p, q\} = 1$ everywhere and all the ρ -norms of the paper are genuine norms.¹

2. NOTATION

We removed from the paper’s table the rows for Hölder atoms, bounded variation atoms, the constants ρ and $\hat{t}$ (no longer needed in the Banach range) since the corresponding material is not formalized. Constants denoted C , $C(\cdot)$ below depend only on the displayed arguments; the Lean statements carry explicit values for most of them.

¹In Lean the standing hypotheses are $1 \leq p, p \neq \infty$, $\text{Fact } (1 \leq q), 1 \leq u, 0 < s$.

TABLE 1. Most important notation/symbols/constants

Symbol	Description
I	phase space
m	finite measure in the phase space I
a_P, b_P	an atom supported on P
$\mathcal{A}$	a class of atoms
$\mathcal{A}(Q)$	set of atoms of class $\mathcal{A}$ supported on Q
$\mathcal{A}_{s,p}^{sz}$	class of (s, p) -Souza's atoms
$\mathcal{A}_{s,\beta,p,q}^{bs}$	class of (s, β, p, q) -Besov's atoms
$\mathcal{P}$	grid of I
$\mathcal{P}^k$	family of subsets of I at the k -th level of $\mathcal{P}$
$\lambda_1 \leq \lambda_2$	describes geometry of the good grid $\mathcal{P}$
$\mathcal{B}_{p,q}^s(\mathcal{A})$	(s, p, q) -Banach space defined by the class of atoms $\mathcal{A}$
$\mathcal{B}_{p,q}^s$ or $\mathcal{B}_{p,q}^s(\mathcal{A}_{s,p}^{sz})$	(s, p, q) -Banach space defined by Souza's atoms
P, Q, W	elements of the grid $\mathcal{P}$
L^p or $L^p(m)$	Lebesgue spaces of $(I, \mathbb{A}, m)$
$ \cdot _p$	norm in L^p , $p \in (0, \infty]$
p'	$1/p + 1/p' = 1$, with $p \in [1, \infty]$

I. DIVIDE AND RULE

In Part I we assume $s > 0$, $p \in [1, \infty)$, $u \in [1, \infty]$ and $q \in [1, \infty]$.^a

^aPaper: $s > 0$, $p \in (0, \infty)$, $q \in (0, \infty]$. See the Introduction for the restriction to the Banach range.

3. MEASURE SPACES AND GRIDS

Let I be a measure space with a σ -algebra $\mathbb{A}$ and m be a measure on $(I, \mathbb{A})$, $m(I) < \infty$. Given a measurable set $J \subset I$ denote $|J| = m(J)$. We denote the Lebesgue spaces of $(I, \mathbb{A}, m)$ by L^p . A **grid** is a sequence of finite families of measurable sets with positive measure $\mathcal{P} = (\mathcal{P}^k)_{k \in \mathbb{N}}$, so that at least one of these families is non empty and

G_1 . Given $Q \in \mathcal{P}^k$, let

$$\Omega_Q^k = \{P \in \mathcal{P}^k : P \cap Q \neq \emptyset\}.$$

Then

$$C_1 = \sup_k \sup_{Q \in \mathcal{P}^k} \#\Omega_Q^k < \infty.$$

Define $|\mathcal{P}^k| = \sup\{|Q| : Q \in \mathcal{P}^k\}$. We often abuse notation using $\mathcal{P}$ for both $(\mathcal{P}^k)_{k \in \mathbb{N}}$ and $\cup_k \mathcal{P}^k$.²

```

structure WeakGrid where
  /-- The measure on ' $\alpha$ '. -/
   $\mu$  : MeasureTheory.Measure  $\alpha$ 
  /-- The measure is finite on the whole space. -/
  isFinite : MeasureTheory.IsFiniteMeasure  $\mu$ 
  /-- The finite family ' $\mathcal{P}^k$ ' at each level ' $k$ '. -/
  partitions :  $\mathbb{N} \rightarrow \text{Finset (Set } \alpha \text{)}$ 
  /-- Every cell is measurable. -/
  measurable :  $\forall k Q, Q \in \text{partitions } k \rightarrow \text{MeasurableSet } Q$ 
  /-- Every cell has positive measure. -/
  positive_measure :  $\forall k Q, Q \in \text{partitions } k \rightarrow 0 < \mu Q$ 
  /-- At least one level is nonempty. -/
  exists_nonempty :  $\exists k, (\text{partitions } k). \text{Nonempty}$ 
  /-- Uniform same-level overlap multiplicity bound. -/
  Cmult1 :  $\mathbb{N}$ 
  /-- ' $\# \Omega_Q^k \leq Cmult1$ ' for every ' $Q \in \mathcal{P}^k$ '. -/
  overlap_card_le :
     $\forall k Q, Q \in \text{partitions } k \rightarrow (\text{overlapFinset (partitions } k) Q). \text{card} \leq Cmult1$ 

structure WeakGridSpace where
  grid : WeakGrid ( $\alpha := \alpha$ )

```

Remark 3.1. There are plenty of examples of spaces with grids; the simplest one is $[0, 1)$ with the Lebesgue measure and the dyadic grid $\mathcal{D}^k = \{[i/2^k, (i+1)/2^k), 0 \leq i < 2^k\}$, and similarly the dyadic and d -adic grids of $[0, 1)^n$. Any measure space with a finite non-atomic measure can be endowed with a grid made of a nested sequence of partitions whose elements at the same level have precisely the same measure.

4. ATOMS

Let $\mathcal{P}$ be a grid. Let $p \in [1, \infty)$, $u \in [1, \infty]$, and $s > 0$. A family of **atoms** associated with $\mathcal{P}$ of type (s, p, u) is an indexed family $\mathcal{A}$ of pairs $(\mathcal{B}(Q), \mathcal{A}(Q))_{Q \in \cup_k \mathcal{P}^k}$ where

²Lean: `structure WeakGrid (fields partitions, measurable, positive_measure, exists_nonempty, Cmult1, overlap_card_le) and WeakGridSpace, in WeakGrid/Definition.lean`. The paper calls this simply a “grid”; in the repository it is called a *weak grid*, to distinguish it from the good grids of Section 8. The paper’s convention that $P \neq Q$ for cells at different levels is replaced by indexing cells by (level, cell) pairs (`WeakGridCell`).

- A₁. $\mathcal{B}(Q)$ is a complex linear space of functions contained in L^{pu} .³
- A₂. If $\phi \in \mathcal{B}(Q)$ then $\phi(x) = 0$ for every $x \notin Q$.
- A₃. $\mathcal{A}(Q)$ is a nonempty convex subset of $\mathcal{B}(Q)$ such that $\phi \in \mathcal{A}(Q)$ implies $\sigma\phi \in \mathcal{A}(Q)$ for every $\sigma \in \mathbb{C}$ satisfying $|\sigma| = 1$.⁴
- A₄. We have

$$|\phi|_{pu} \leq |Q|^{s - \frac{1}{u'p}}$$

for every $\phi \in \mathcal{A}(Q)$, where $1/u + 1/u' = 1$.

We will say that $\phi \in \mathcal{A}(Q)$ is an $\mathcal{A}$ -atom of type (s, p, u) supported on Q . Sometimes we also assume

- A₅. For every $Q \in \mathcal{P}$ we have that $\mathcal{A}(Q)$ is a compact subset in the strong topology of L^p ,

or

- A₆. every $\mathcal{A}(Q)$ is a sequentially compact subset in the weak topology of L^p .⁵

```

structure AtomFamily
  (G : WeakGridSpace (α := α)) (s : ℝ) (p u : ℝ ≥ 0 ∞) where
    /-- The Hölder conjugate exponent 'u'. -/
    uConj : ℝ ≥ 0 ∞
    /-- The smoothness parameter is positive. -/
    s_pos : 0 < s
    /-- 'p ∈ [1, ∞)'. -/
    one_le_p : 1 ≤ p
    /-- 'p < ∞', expressed in 'ℝ ≥ 0 ∞' as 'p ≠ ∞'. -/
    p_ne_top : p ≠ ∞
    /-- 'u ∈ [1, ∞]'. -/
    one_le_u : 1 ≤ u
    /-- 'uConj' is the Hölder conjugate of 'u'. -/
    holder_conjugate : ENNReal.HolderConjugate u uConj
    /-- The local vector space 'B(Q)'. -/
    localSpace : WeakGridCell G → LocalVectorSpace.{u, v} α
    /-- The chosen atoms 'A(Q)', as elements of 'B(Q)'. -/
    atoms : ∀ Q, Set ((localSpace Q).carrier)
    /-- Every cell has at least one atom. -/
    atoms_nonempty : ∀ Q, (atoms Q).Nonempty
    /-- 'B(Q)' is contained in 'L^{pu}'. -/

```

³Paper A₁ asks $\mathcal{B}(Q)$ to be a complex *Banach* space contained in L^{pu} . Lean (`LocalVectorSpace`, in `WeakGrid/Atoms.lean`) only requires a complex vector space together with an injective linear map into the functions $\alpha \rightarrow \mathbb{C}$, all whose members belong to L^{pu} (`local_memLp`); no completeness of the local space is assumed, and it is never used.

⁴Lean: `atoms_nonempty`, `atoms_convex`, `atoms_phase_invariant`. The paper states the phase invariance as an “if and only if”; the two formulations are equivalent.

⁵Lean: `structure AtomFamily` in `WeakGrid/Atoms.lean`; A₅ appears as `AssumptionA5`. The paper’s alternative assumption A₇ ($\mathcal{B}(Q)$ finite dimensional and $\mathcal{A}(Q)$ closed) is used in the paper only to derive A₅; in the formalization the finite-dimensional case is instantiated directly for Souza’s atoms.

```

local_memLp : ∀ Q φ, MeasureTheory.MemLp ((localSpace Q).toFun φ) (p * u)
  G.measure
/-- Local functions are supported on 'Q'. -/
local_support : ∀ Q φ, ∀ x, x ∉ Q.cell → (localSpace Q).toFun φ x = 0
/-- 'A(Q)' is convex in the local vector space. -/
atoms_convex : ∀ Q, Convex ℝ (atoms Q)
/-- 'A(Q)' is invariant under multiplication by complex scalars of modulus one. -/
atoms_phase_invariant :
  ∀ Q φ (σ : ℂ), φ ∈ atoms Q → ||σ|| = (1 : ℝ) → σ • φ ∈ atoms Q
/-- The atom size estimate in 'L^{pu}'. -/
atom_bound :
  ∀ Q φ, φ ∈ atoms Q →
    MeasureTheory.eLpNorm ((localSpace Q).toFun φ) (p * u) G.measure ≤
      atomMeasureScale G s p uConj Q

```

5. BESOV-ISH SPACES

Let $p \in [1, \infty)$, $u \in [1, \infty]$, $q \in [1, \infty]$, $s > 0$, $\mathcal{P} = (\mathcal{P}^k)_{k \geq 0}$ be a grid and let $\mathcal{A}$ be a family of atoms of type (s, p, u) . We will also assume

G_2 . We have

$$\left(\sum_k |\mathcal{P}^k|^{sq^*} \right)^{1/q^*} < \infty \quad (\text{usual modification if } q^* = \infty), \quad \lim_k |\mathcal{P}^k| = 0,$$

where q^* is the exponent produced by the Hölder-like estimate of the paper (for $q > 1$, $q^* = q'$).⁶

The **Besov-ish space** $\mathcal{B}_{p,q}^s(I, \mathcal{P}, \mathcal{A})$ is the set of all complex valued functions $g \in L^p$ that can be represented by an absolutely convergent series in L^p

$$(1) \quad g = \sum_{k=0}^{\infty} \sum_{Q \in \mathcal{P}^k} s_Q a_Q$$

where a_Q is in $\mathcal{A}(Q)$, $s_Q \in \mathbb{C}$ and with finite **cost**

$$(2) \quad \left(\sum_{k=0}^{\infty} \left(\sum_{Q \in \mathcal{P}^k} |s_Q|^p \right)^{q/p} \right)^{1/q} < \infty$$

(usual modification when $q = \infty$). The series in (1) is called a $\mathcal{B}_{p,q}^s(I, \mathcal{P}, \mathcal{A})$ -**representation** of the function g . Define

$$|g|_{\mathcal{B}_{p,q}^s(I, \mathcal{P}, \mathcal{A})} = \inf \left(\sum_{k=0}^{\infty} \left(\sum_{Q \in \mathcal{P}^k} |s_Q|^p \right)^{q/p} \right)^{1/q},$$

⁶Lean: `AssumptionG2` in `WeakGrid/BesovishSpaces.lean`. The paper writes this condition as $C(p, q, (|\mathcal{P}^k|^s)_k) < \infty$ using the constant of its Proposition 4.1; since Section 4 has no standalone Lean counterpart, the formalization states the required summability of the weighted level measures directly, together with $\lim_k |\mathcal{P}^k| = 0$.

where the infimum runs over all possible representations of g as in (1).⁷ When the choice of I and/or $\mathcal{P}$ is obvious we write $\mathcal{B}_{p,q}^s(\mathcal{P}, \mathcal{A})$ or $\mathcal{B}_{p,q}^s(\mathcal{A})$; whenever we write just $\mathcal{B}_{p,q}^s$ we choose the Souza's atoms $\mathcal{A}_{s,p}^{sz}$ of Section 10.

```

structure LpGridRepresentation
  (A : AtomFamily G s p u) (g : Lp C p G.measure) where
  block : (k : ℕ) → LevelBlock A k
  hasSum : HasSum (fun k => (block k).toLp A) g

def BesovishSpace (A : AtomFamily G s p u) (q : ℝ ≥ 0∞)
  [Fact (1 ≤ q)]
  : Submodule C (Lp C p G.measure) where
  -- Carrier: all 'L^p' elements admitting a Besov-ish atomic
  -- representation with finite p q cost.
  carrier := { g | MemBesovishCoeffCost A q g }
  ...

noncomputable def Norm_Costpq
  (A : AtomFamily G s p u) (q : ℝ ≥ 0∞) [Fact (1 ≤ q)]
  (g : BesovishSpace A q) : ℝ :=
  pqPseudoNorm A q g

```

Proposition 5.1 (Embedding into L^t). *Assume G_1 – G_2 and A_1 – A_4 . Let $t \in [p, \infty)$ be such that*

$$(3) \quad s - \frac{1}{p} + \frac{1}{t} \geq 0, \quad p \leq t \leq pu,$$

and suppose

$$(4) \quad C_2(t) = C_1^{1+1/t} C(t, q, (|\mathcal{P}^k|^{t(s-\frac{1}{p}+\frac{1}{t})})_k) < \infty.$$

Then for every g with a $\mathcal{B}_{p,q}^s(\mathcal{A})$ -representation (1) of cost K the series converges in L^t and

$$(5) \quad |g|_t \leq C_2(t) K.$$

In particular

$$|g|_t \leq C_2(t) |g|_{\mathcal{B}_{p,q}^s(\mathcal{A})},$$

*and $\mathcal{B}_{p,q}^s(\mathcal{A}) \subset L^t$ continuously.*⁸

⁷Lean: `LpGridRepresentation` (a representation is a sequence of level blocks together with a `HasSum` witness in L^p), `pqCost`, `MemBesovishCoeffCost`, `BesovishSpace` (a `Submodule` of L^p) and the cost norm `Norm_Costpq` (an infimum over representation costs), all in `WeakGrid/BesovishSpaces.lean`.

⁸Lean: `lp_embedding_adapted_statement` in `WeakGrid/BesovishSpaces.lean`. The Lean statement is the per-representation inequality (5) with the explicit constant $C_1^{1+1/t} \cdot \text{cCoefficient}$; the norm inequality follows by taking the infimum over representations. Compared with paper Proposition 6.1 the hypotheses are identical except for

```

theorem lp_embedding_adapted_statement
  {A : AtomFamily G s p u} {t : ℝ ≥ 0 ∞}
  [Fact (1 ≤ t)]
  (hp_ne_top : p ≠ ∞) (ht_ne_top : t ≠ ∞)
  (hq_one : 1 ≤ q)
  (hp_le_t : p ≤ t) (ht_le_pu : t ≤ p * u)
  (hs_nonneg : 0 ≤ s - 1 / p.toReal + 1 / t.toReal)
  {g : Lp ℂ p G.measure} (R : LpGridRepresentation A g)
  (hRfin : LpGridRepresentation.FinitePQCost (q := q) R)
  (hCco_fin : cCoefficientFinite t q (fun k =>
    (levelMeasureWeight G s p t k) ^ t.toReal)) :
    (MeasureTheory.eLpNorm (g : α → ℂ) t G.measure).toReal ≤
    ((G.grid.Cmult1 : ℝ) ^ (1 + 1 / t.toReal)) *
    cCoefficient t q (fun k => (levelMeasureWeight G s p t k) ^
    t.toReal) *
    LpGridRepresentation.pqCost (q := q) R := ...

```

Proposition 5.2 (Linear structure). *Assume G_1 – G_2 and A_1 – A_4 . Then $\mathcal{B}_{p,q}^s(\mathcal{A})$ is a complex linear subspace of L^p and $|\cdot|_{\mathcal{B}_{p,q}^s(\mathcal{A})}$ is a norm on it. Moreover the inclusion*

$$\iota : (\mathcal{B}_{p,q}^s(\mathcal{A}), |\cdot|_{\mathcal{B}_{p,q}^s(\mathcal{A})}) \rightarrow (L^p, |\cdot|_p)$$

*is continuous.*⁹

```

noncomputable def costNormedAddCommGroup
  (hp_top : p ≠ ∞)
  (hCco_fin : LpGridRepresentation.cCoefficientFinite p q (fun k =>
    (LpGridRepresentation.levelMeasureWeight G s p p k) ^ p.toReal)) :
  NormedAddCommGroup (BesovishSpace A q) where
  norm := Norm_Costpq A q
  dist x y := Norm_Costpq A q (-x + y)
  ...

```

Proposition 5.3 (Limits of representations). *Assume G_1 – G_2 and A_1 – A_4 . Suppose that g_n are functions in $\mathcal{B}_{p,q}^s(\mathcal{A})$ with $\mathcal{B}_{p,q}^s(\mathcal{A})$ -representations*

$$g_n = \sum_{k=0}^{\infty} \sum_{Q \in \mathcal{P}^k} s_Q^n a_Q^n$$

satisfying

$q \geq 1$ and $t \geq p$ being imposed globally, and the levelwise estimate (6.2) of the paper is internal to the proof rather than part of the statement.

⁹Lean: `BesovishSpace` is a `Submodule` of L^p ; the normed structure is `BesovishSpace.costNormedAddCommGroup` (requiring $p \neq \infty$ and G_2), and continuity of ι is the case $t = p$ of Proposition 5.1. Compared with paper Proposition 6.3, $\rho = \min\{1, p, q\} = 1$ since $p, q \geq 1$, so the ρ -norm of the paper is a norm.

i. There is C such that for every n

$$\left(\sum_{k=0}^{\infty} \left(\sum_{Q \in \mathcal{P}^k} |s_Q^n|^p \right)^{q/p} \right)^{1/q} \leq C.$$

ii. For every $Q \in \mathcal{P}$ the limit $s_Q = \lim_n s_Q^n$ exists.

iii. For every $Q \in \mathcal{P}$ there is $a_Q \in \mathcal{A}(Q)$ such that either $a_Q^n \rightarrow a_Q$ in the strong topology of L^p , or $a_Q^n \rightarrow a_Q$ weakly in L^p .

Then g_n converges (strongly, respectively weakly) in L^p to a function $g \in \mathcal{B}_{p,q}^s(\mathcal{A})$ which has the $\mathcal{B}_{p,q}^s(\mathcal{A})$ -representation

$$g = \sum_{k=0}^{\infty} \sum_{Q \in \mathcal{P}^k} s_Q a_Q$$

of cost at most C .¹⁰

```

theorem representation_limit
  (A : AtomFamily G s p u) (hG2 : AssumptionG2 G s p u q)
  {gseq : ℕ → Lp ℂ p G.measure}
  {gLim : Lp ℂ p G.measure} {C : ℝ}
  (H : RepresentationLimitHypotheses A q gseq gLim C)
  (hp_ne_top : p ≠ ∞) (hs_pos : 0 < s) (hu_one : 1 ≤ u)
  [Fact (1 ≤ u)] (hC : 0 ≤ C) :
  MemBesovishCoeffCost A q gLim ∧
  LpGridRepresentation.FinitePQCost (q := q) H.Rlim ∧
  LpGridRepresentation.pqCost (q := q) H.Rlim ≤ C ∧
  Tendsto (fun n => toWeakSpace ℂ (Lp ℂ p G.measure) (gseq n)) atTop
  (N (toWeakSpace ℂ (Lp ℂ p G.measure) gLim)) := ...

```

Corollary 5.4 (Compactness and completeness). *Assume G_1 – G_2 , A_1 – A_4 , and either A_5 or A_6 . Then*

- i. *If $g_n \in \mathcal{B}_{p,q}^s(\mathcal{A})$ satisfy $|g_n|_{\mathcal{B}_{p,q}^s(\mathcal{A})} \leq C$ for every n , then there is a subsequence converging (strongly under A_5 , weakly under A_6) in L^p to some $g \in \mathcal{B}_{p,q}^s(\mathcal{A})$ with $|g|_{\mathcal{B}_{p,q}^s(\mathcal{A})} \leq C$.¹¹*
- ii. *Under A_5 , every closed ball of $(\mathcal{B}_{p,q}^s(\mathcal{A}), |\cdot|_{\mathcal{B}_{p,q}^s(\mathcal{A})})$ is sequentially compact in the strong topology of L^p .¹²*

¹⁰Lean: `representation_limit` and `representation_limit_strong` in `WeakGrid/Completeness.lean`, matching paper Proposition 6.4.

¹¹Lean: `representation_limit_strong_existence` and `representation_limit_weak_existence` in `WeakGrid/Completeness.lean` (paper Corollary 6.5).

¹²Lean: `exists_strongly_convergent_subseq_of_uniform_pqCost` and `closed_Norm_Costpq_ball_strongly_seqCompact` in `WeakGrid/Completeness.lean` (paper Corollary 6.6; the paper states i. and the compactness of the inclusion ι , the

iii. Under A_5 , $(\mathcal{B}_{p,q}^s(\mathcal{A}), |\cdot|_{\mathcal{B}_{p,q}^s(\mathcal{A})})$ is a complex Banach space.¹³

```

theorem closed_Norm_Costpq_ball_strongly_seqCompact
  (hp_ne_top : p ≠ ∞) (hs_pos : 0 < s) (hu_one : 1 ≤ u)
  (A : AtomFamily G s p u) (hG2 : AssumptionG2 G s p u q)
  (hA5 : AssumptionA5 A)
  {C : ℝ}
  [Fact (1 ≤ u)]
  (hC : 0 ≤ C)
  (gseq : ℕ → BesovishSpace A q)
  (hball : ∀ n, BesovishSpace.Norm_Costpq A q (gseq n) ≤ C) :
  ∃ (φ : ℕ → ℕ) (_hφ : StrictMono φ) (gLim : BesovishSpace A q),
    BesovishSpace.Norm_Costpq A q gLim ≤ C ∧
    Tendsto
      (fun n => ((gseq (φ n) : BesovishSpace A q) : Lp ℂ p G.measure))
      atTop
      (λ (gLim : BesovishSpace A q) : Lp ℂ p G.measure) := ...

theorem besovishSpace_costNorm_completeSpace
  (hp_ne_top : p ≠ ∞) (hs_pos : 0 < s) (hu_one : 1 ≤ u)
  (A : AtomFamily G s p u) (hG2 : AssumptionG2 G s p u q)
  (hA5 : AssumptionA5 A)
  [Fact (1 ≤ u)] :
  @CompleteSpace (BesovishSpace A q)
    (BesovishSpace.costNormedAddCommGroup
      (A := A) (q := q) hp_ne_top
      hG2.1).toMetricSpace.toPseudoMetricSpace.toUniformSpace := ...

```

6. SCALES OF SPACES

A family of atoms $\mathcal{A}$ of type (s, p, u) induces a one-parameter scale $\tilde{s} \mapsto \mathcal{A}_{\tilde{s},p}$, where $\mathcal{A}_{\tilde{s},p}$ is the family of atoms of type $(\tilde{s}, p, u)$ defined by

$$\mathcal{A}_{\tilde{s},p}(Q) = \{|Q|^{\tilde{s}-s} a_Q : a_Q \in \mathcal{A}(Q)\}.$$

Proposition 6.1 (Scale inclusion). *Assume G_1 – G_2 and that $\mathcal{A}$ satisfies A_1 – A_4 . Let $0 < s \leq \tilde{s}$ and $q, \tilde{q} \in [1, \infty]$. Suppose*

$$\left(\sum_k |\mathcal{P}^k|^{q^*(\tilde{s}-s)} \right)^{1/q^*} < \infty.$$

formalization states the sequential compactness of closed cost balls, which is the form used downstream).

¹³Lean: `besovishSpace_costNorm_completeSpace` in `WeakGrid/Completeness.lean` (paper Corollary 6.7; for $p, q \geq 1$ the ρ -Banach space of the paper is a Banach space). The paper's Corollary 6.6 (existence of a representation attaining the infimum cost) is not formalized and is omitted.

Then $\mathcal{B}_{p,\tilde{q}}^{\tilde{s}}(\mathcal{A}_{\tilde{s},p}) \subset \mathcal{B}_{p,q}^s(\mathcal{A}_{s,p})$, the inclusion is a continuous linear map,¹⁴ and

$$|g|_{\mathcal{B}_{p,q}^s(\mathcal{A}_{s,p})} \leq C(q, (|\mathcal{P}^k|^{\tilde{s}-s})_k) |g|_{\mathcal{B}_{p,\tilde{q}}^{\tilde{s}}(\mathcal{A}_{\tilde{s},p})}.$$

```

theorem smoothnessScaleBesovishSpace_subset
[Fact (1 ≤ p)]
[Fact (1 ≤ u)] [Fact (1 ≤ q)] [Fact (1 ≤ qTilde)]
(A : AtomFamily G s p u) (sTilde : ℝ) (hsTilde_pos : 0 < sTilde)
(hs_le : s ≤ sTilde)
(hqfin : scaleCoefficientFinite q (smoothnessScaleLevelWeight G s
sTilde p))
{g : Lp ℂ p G.measure}
(hg : g ∈ SmoothnessScaleBesovishSpace
(A.smoothnessScaleAtomFamily sTilde hsTilde_pos) qTilde) :
g ∈ BesovishSpace A q := ...

```

7. TRANSMUTATION OF ATOMS

The key result in Part I states that if we replace each atom of an atomic decomposition by a combination of atoms of a possibly different class, in such way that we do not change the location of the supports and the mass of the new representation concentrates pretty much on the original scale, then we obtain a new atomic decomposition whose cost is controlled by the cost of the original one.

Proposition 7.1 (Transmutation of atoms). *Assume*

- I. $\mathcal{A}_2$ is a class of (s, p, u_2) -atoms for a grid $\mathcal{W}$, satisfying A_1 – A_4 and G_1 – G_2 . Let $\mathcal{G}$ be also a grid satisfying G_1 – G_2 .
- II. $k_i \in \mathbb{N}$, $i \geq 0$, is a sequence such that there are $\alpha > 0$ and $A, B \in \mathbb{R}$ with

$$\alpha i + A \leq k_i \leq \alpha i + B \quad \text{for every } i.$$

- III. There is $\lambda \in (0, 1)$ such that the following holds. For every $Q \in \mathcal{G}^i$ there are atoms $b_{P,Q} \in \mathcal{A}_2(P)$ and coefficients $s_{P,Q} \in \mathbb{C}$, indexed by the cells $P \in \mathcal{W}$ with $P \subset Q$, such that

$$h_Q = \sum_k \sum_{P \in \mathcal{W}^k, P \subset Q} s_{P,Q} b_{P,Q}$$

¹⁴Lean: `smoothnessScaleBesovishSpace_subset` and `smoothnessScaleBesovishSpaceInclusion.Norm.Costpq_le` in `WeakGrid/Scales.lean`, with the explicit constant `scaleCoefficient`. This is item A of paper Proposition 7.1; items B (compactness of bounded sets of the finer space in the coarser norm) and C (compactness of the inclusion under A_7) are not formalized and are omitted. The paper's two-parameter scale $(\tilde{s}, \tilde{p})$ is also not formalized.

is a $\mathcal{B}_{p,q}^s(\mathcal{A}_2)$ -representation of a function h_Q , with $s_{P,Q} = 0$ whenever $P \in \mathcal{W}^k$ with $k < k_i$, and moreover

$$(6) \quad \sum_{P \in \mathcal{W}^k, P \subset Q} |s_{P,Q}|^p \leq C_3 \lambda^{k-k_i} \quad \text{for every } k \geq k_i.$$

Then

A. For all coefficients $(c_Q)_{Q \in \mathcal{G}}$ with

$$\left(\sum_i \left(\sum_{Q \in \mathcal{G}^i} |c_Q|^p \right)^{q/p} \right)^{1/q} < \infty,$$

the sequence $N \mapsto \sum_{i \leq N} \sum_{Q \in \mathcal{G}^i} c_Q h_Q$ converges in L^p to a function in $\mathcal{B}_{p,q}^s(\mathcal{A}_2)$ that has a $\mathcal{B}_{p,q}^s(\mathcal{A}_2)$ -representation $\sum_k \sum_P m_P d_P$ with $m_P \geq 0$ and

$$\left(\sum_k \left(\sum_{P \in \mathcal{W}^k} |m_P|^p \right)^{q/p} \right)^{1/q} \leq C_4 \left(\sum_i \left(\sum_{Q \in \mathcal{G}^i} |c_Q|^p \right)^{q/p} \right)^{1/q},$$

where $C_4 = C_4(C_1, C_3, \lambda, \alpha, A, B, p, q)$ is explicit.¹⁵

- B.** Suppose, in addition to the assumptions of A., that the $s_{P,Q}$ are nonnegative real numbers and $b_{P,Q} > 0$ on P for all P, Q . If $m_P \neq 0$ and $d_P \neq 0$ somewhere on P , then $P \subset Q$ for some cell $Q \in \mathcal{G}$ with $c_Q \neq 0$ and $s_{P,Q} > 0$, and the corresponding h_Q is a positive multiple of a positive function a.e. on P . If moreover $c_Q \geq 0$ for every Q , then $m_P \neq 0$ implies $d_P > 0$ a.e. on P .¹⁶
- C.** Let $\mathcal{A}_1$ be a class of (s, p, u_1) -atoms for the grid $\mathcal{G}$ satisfying A_1 – A_4 . Suppose that for every atom $a_Q \in \mathcal{A}_1(Q)$ we can find $s_{P,Q}$ and $b_{P,Q}$ as in III. such that $h_Q = a_Q$. Then

$$\mathcal{B}_{p,q}^s(\mathcal{A}_1) \subset \mathcal{B}_{p,q}^s(\mathcal{A}_2)$$

and this inclusion is continuous, with $|\phi|_{\mathcal{B}_{p,q}^s(\mathcal{A}_2)} \leq C_4 |\phi|_{\mathcal{B}_{p,q}^s(\mathcal{A}_1)}$ for every $\phi \in \mathcal{B}_{p,q}^s(\mathcal{A}_1)$.¹⁷

```
theorem Transmutation_of_Atoms_Claim_A
  (G W : WeakGridSpace (α := α))
  (AW : AtomFamily W s p u)
```

¹⁵Lean: `Transmutation_of_Atoms_Claim_A` in `WeakGrid/Transmutation.lean`, with the explicit constant of the paper (product of C_1 , $C_3^{1/p}$, $\lambda^{-B/p}$, the convolution-trick coefficient and the integer-part factor $\max\{\ell \in \mathbb{N} : \ell < \alpha\} + 1$).

¹⁶Lean: `Transmutation_of_Atoms_Claim_B` and `Transmutation_of_Atoms_Claim_B_sharp` in `WeakGrid/Transmutation.lean`. Compared with paper Proposition 8.1.B, the pointwise positivity conclusions are stated almost everywhere on P (with positivity witnessed by positive real multiples), which is the natural formulation in L^p .

¹⁷Lean: `Transmutation_of_Atoms_Claim_C_explicit` and `Transmutation_of_Atoms_continuous_embedding_explicit` in `WeakGrid/Transmutation.lean`.

```

(k : ℕ → ℕ)
(A_als B_als r_als : ℝ)
(hr_als : 0 < r_als)
(hk_bound : ∀ i : ℕ,
  (k i : NNReal) ≤ r_als * (i : NNReal) + B_als ∧
  r_als * (i : NNReal) + A_als ≤ (k i : NNReal))
(lam : ℝ) (hlam_pos : 0 < lam) (hlam_lt : lam < 1)
(C : ℝ) (hC : 0 ≤ C)
(h : (i : ℕ) → LevelCell G i → Lp ℂ p W.measure)
(R : (i : ℕ) → (Q : LevelCell G i) → LpGridRepresentation AW (h i Q))
(hR : RepresentationWsubGandALS (p := p) (q := q) G W AW k
  ⟨A_als, B_als, r_als, hr_als, hk_bound⟩ lam hlam_pos hlam_lt C hC h R)
(c : (i : ℕ) → LevelCell G i → ℂ)
(hc : CoeffFinitePQCost (p := p) (q := q) G c)
(hq_ne_top : q ≠ ∞)
(hG2_W : AssumptionG2 W s p u q)
(hp_ne_top : p ≠ ∞)
(hs_pos : 0 < s) :
∃ gLim : Lp ℂ p W.measure,
  ∃ hsum :
    HasSum (fun j => (TransmutationBlockLimit G W AW h R c A_als r_als j).toLp
      AW) gLim,
    MemBesovishCoeffCost AW q gLim ∧
    LpGridRepresentation.FinitePQCost (q := q)
      ({ block := TransmutationBlockLimit G W AW h R c A_als r_als
        hasSum := hsum } : LpGridRepresentation AW gLim) ∧
    Tendsto (fun N => PartialSumLevels G W h c N) atTop (λ gLim) ∧
    LpGridRepresentation.pqCost (q := q)
      ({ block := TransmutationBlockLimit G W AW h R c A_als r_als
        hasSum := hsum } : LpGridRepresentation AW gLim) ≤
      (G.grid.Cmulti : ℝ) *
      C ^ (1 / p.toReal) *
      lam ^ (-(B_als : ℝ) / p.toReal) *
      LpGridRepresentation.cCoefficientInt p ∞
      (transmutationKernelZ lam A_als r_als) *
      (Nat.ceil (r_als : ℝ) : ℝ) ^ (1 / q.toReal) *
      CoeffPQCost (p := p) (q := q) G c := ...

theorem Transmutation_of_Atoms_Claim_B (G W : WeakGridSpace (α := α))
  (AW : AtomFamily W s p u)
  (k : ℕ → ℕ)
  (A_als B_als r_als : ℝ)
  (hr_als : 0 < r_als)
  (hk_bound : ∀ i : ℕ,
    (k i : NNReal) ≤ r_als * (i : NNReal) + B_als ∧
    r_als * (i : NNReal) + A_als ≤ (k i : NNReal))
  (lam : ℝ) (hlam_pos : 0 < lam) (hlam_lt : lam < 1)
  (C : ℝ) (hC : 0 ≤ C)
  (h : (i : ℕ) → LevelCell G i → Lp ℂ p W.measure)
  (R : (i : ℕ) → (Q : LevelCell G i) → LpGridRepresentation AW (h i Q))
  (hR : RepresentationWsubGandALS_pos (p := p) (q := q) G W AW k
    ⟨A_als, B_als, r_als, hr_als, hk_bound⟩ lam hlam_pos hlam_lt C hC h R)
  (c : (i : ℕ) → LevelCell G i → ℂ)
  (hc : CoeffFinitePQCost (p := p) (q := q) G c)
  (hG2_W : AssumptionG2 W s p u q)

```

```

(hp_ne_top : p ≠ ∞)
(hs_pos : 0 < s) :
(∃ gLim : Lp ℂ p W.measure,
  HasSum (fun j => (TransmutationBlockLimit G W AW h R c A_als r_als j).toLp
    AW) gLim ∧
  MemBesovishCoeffCost AW q gLim ∧
  Tendsto (fun N => PartialSumLevels G W h c N) atTop (λ gLim) ∧
  (∀ j : ℕ, ∀ P : LevelCell W j,
    TransmutationCoeffLimit G W AW h R c A_als r_als P ≠ 0 →
    (∃ x, x ∈ P.1 →
      AW.toFunction (levelCellToWeakGridCell W j P)
        (TransmutationAtomLocalLimit G W AW h R c A_als r_als P) x ≠ 0) →
    ∃ i ∈ Finset.range (transmutationStabilizationIndex A_als r_als j),
    ∃ Q ∈ (G.grid.partitions i).attach,
    P.1 ⊆ Q.1 ∧ c i Q ≠ 0 ∧
    ∃ r : NNReal, 0 < r ∧ ((R i Q).block j).coeff P = (r : ℂ) ∧
    ∀m x ∂ W.measure.restrict P.1,
    ∃ r : NNReal, 0 < r ∧ h i Q x = (r : ℂ)) := ...)

theorem Transmutation_of_Atoms_continuous_embedding_explicit
  (G : WeakGridSpace (α := α))
  (u1 u2 : ℝ≥0∞)
  [Fact (1 ≤ u2)]
  (AG1 : AtomFamily G s p u1)
  (AG2 : AtomFamily G s p u2)
  (lam : ℝ) (hlam_pos : 0 < lam) (hlam_lt : lam < 1)
  (C : ℝ) (hC : 0 ≤ C)
  (hAG1_to_AG2 : ∀ i : ℕ, ∀ Q : LevelCell G i,
    ∀ g : Lp ℂ p G.measure,
    (∃ φ : (AG1.localSpace (levelCellToWeakGridCell G i Q)).carrier,
      AG1.IsAtom (levelCellToWeakGridCell G i Q) φ ∧
      g = atomLp AG1 (levelCellToWeakGridCell G i Q) φ) →
    ∃ Rg : LpGridRepresentation AG2 g,
      CoeffFinitePQCost (p := p) (q := q) G
        (fun j S => (Rg.block j).coeff S) ∧
      (∀ j : ℕ, ∀ S : LevelCell G j,
        (¬ S.1 ⊆ Q.1 → (Rg.block j).coeff S = 0) ∧
        (j < i → (Rg.block j).coeff S = 0)) ∧
      ∀ j : ℕ, i ≤ j → Rg.levelCoeffPower j ≤ C * lam ^ (j - i))
  (hg2_G : AssumptionG2 G s p u2 q)
  (hp_ne_top : p ≠ ∞)
  (hs_pos : 0 < s) :
0 ≤ transmutationClaimCEmbeddingConstant G p q lam C ∧
  ∀ g : BesovishSpace AG1 q,
  ∃ hg2 : MemBesovishCoeffCost AG2 q (g : Lp ℂ p G.measure),
  BesovishSpace.Norm_Costpq AG2 q
    ((g : Lp ℂ p G.measure), hg2) : BesovishSpace AG2 q ≤
  transmutationClaimCEmbeddingConstant G p q lam C *
  BesovishSpace.Norm_Costpq AG1 q g := ...

```

8. GOOD GRIDS

A (λ_1, λ_2) -**good grid**, with $0 < \lambda_1 \leq \lambda_2 < 1$, is a grid $\mathcal{P} = (\mathcal{P}^k)_{k \in \mathbb{N}}$ with the following properties:

G_3 . $\mathcal{P}^0 = \{I\}$.

G_4 . $I = \cup_{Q \in \mathcal{P}^k} Q$ up to a set of zero m -measure.

G_5 . The elements of $\mathcal{P}^k$ are pairwise disjoint.

G_6 . For every $Q \in \mathcal{P}^k$, $k > 0$, there exists $P \in \mathcal{P}^{k-1}$ such that $Q \subset P$.

G_7 . We have

$$\lambda_1 \leq \frac{|Q|}{|P|} \leq \lambda_2$$

for every $Q \subset P$ with $Q \in \mathcal{P}^{k+1}$ and $P \in \mathcal{P}^k$, $k \geq 0$.

G_8 . The family $\cup_k \mathcal{P}^k$ generates the σ -algebra $\mathbb{A}$.¹⁸

```
structure GoodGrid extends UnbalancedHaarWavelet.Grid (α := α) where
  lambda1 : ℝ
  /-- Upper ratio bound (strictly less than 1). -/
  lambda2 : ℝ
  /-- 'λ₁' is strictly positive. -/
  hlambda1_pos : 0 < lambda1
  /-- 'λ₂' is strictly less than '1'. -/
  hlambda2_lt_one : lambda2 < 1
  /-- 'λ₁ ≤ λ₂'. -/
  hlambda1_le_lambda2 : lambda1 ≤ lambda2
  /-- Each child cell's measure is at least 'λ₁' times its parent's
  measure. -/
  ratio_lower : ∀ (n : ℕ) (s t : Set α),
    s ∈ grid.partitions (n + 1) → t ∈
grid.partitions n → s ⊆ t →
    ENNReal.ofReal lambda1 * μ t ≤ μ s
  /-- Each child cell's measure is at most 'λ₂' times its parent's
  measure. -/
  ratio_upper : ∀ (n : ℕ) (s t : Set α),
    s ∈ grid.partitions (n + 1) → t ∈
grid.partitions n → s ⊆ t →
    μ s ≤ ENNReal.ofReal lambda2 * μ t
```

9. INDUCED SPACES

Consider a Besov-ish space $\mathcal{B}_{p,q}^s(I, \mathcal{P}, \mathcal{A})$, where $\mathcal{P}$ is a good grid. Given $Q \in \mathcal{P}^{k_0}$, consider the sequence of finite families of subsets $\mathcal{P}_Q = (\mathcal{P}_Q^i)_{i \geq 0}$ of

¹⁸Lean: structure GoodGrid in GoodGrid/Definition.lean, which extends the structure UnbalancedHaarWavelet.Grid of the external repository (where the nested-partition properties G_3 – G_6 , G_8 live) with the ratio bounds G_7 (fields `ratio_lower`, `ratio_upper`). The paper requires $\lambda_1 < \lambda_2$; the formalization allows $\lambda_1 \leq \lambda_2$.

Q given by

$$\mathcal{P}_Q^i = \{P \in \mathcal{P}^{k_0+i}, P \subset Q\},$$

and let $\mathcal{A}_Q$ be the restriction of $\mathcal{A}$ to indices in $\mathcal{P}_Q$. Then we can consider the **induced** Besov-ish space $\mathcal{B}_{p,q}^s(Q, \mathcal{P}_Q, \mathcal{A}_Q)$, and the inclusion

$$i: \mathcal{B}_{p,q}^s(Q, \mathcal{P}_Q, \mathcal{A}_Q) \rightarrow \mathcal{B}_{p,q}^s(I, \mathcal{P}, \mathcal{A})$$

is well defined and a weak contraction:¹⁹

$$|f|_{\mathcal{B}_{p,q}^s(I, \mathcal{P}, \mathcal{A})} \leq |f|_{\mathcal{B}_{p,q}^s(Q, \mathcal{P}_Q, \mathcal{A}_Q)}.$$

```
def inducedRepresentationToAmbient
  (G : WeakGridSpace (α := α)) {k₀ : ℕ} (Q : LevelCell G k₀)
  {s : ℝ} {p u : ℝ ≥ 0 ∞} [Fact (1 ≤ p)] (A : AtomFamily G s p u)
  {f : Lp ℂ p (inducedWeakGridSpace G Q).measure}
  (R : LpGridRepresentation (inducedAtomFamily G Q A) f) :
  LpGridRepresentation A (inducedLpToAmbient G Q f) := ...
```

10. EXAMPLES OF CLASSES OF ATOMS: SOUZA'S ATOMS

Let $Q \in \mathcal{P}$. A (s, p) -**Souza's atom** supported on Q is a function $a: I \rightarrow \mathbb{C}$ such that $a(x) = 0$ for every $x \notin Q$ and a is constant on Q , with

$$|a|_\infty \leq |Q|^{s-1/p}.$$

The set of Souza's atoms supported on Q will be denoted by $\mathcal{A}_{s,p}^{sz}(Q)$. The **canonical Souza's atom** on Q is the Souza's atom with $a(x) = |Q|^{s-1/p}$ for every $x \in Q$. Souza's atoms are (s, p, ∞) -type atoms and satisfy A_1 – A_5 .²⁰

```
def IsSouzaAtom (G : GoodGridSpace (α := α)) (s : ℝ) (p : ℝ ≥ 0 ∞)
  (Q : GoodGridCell G) (a : α → ℂ) : Prop :=
  (∀ x ∉ Q.cell, a x = 0) ∧
  ∃ c : ℂ, (∀ x ∈ Q.cell, a x = c) ∧
  ||c|| ≤ (G.grid.μ Q.cell).toReal ^ (s - (p.toReal)⁻¹)
```

II. SPACES DEFINED BY SOUZA'S ATOMS

¹⁹Lean: `inducedWeakGrid`, `inducedWeakGridSpace`, `inducedAtomFamily` and `inducedRepresentationToAmbient` in `WeakGrid/InducedGrid.lean`. The formalization develops the induced grid, induced atoms, the transfer of representations from the induced grid to the ambient one with control of cost, and (for Souza's atoms) the transfer of positivity; this is the form needed in Part III. The reverse restriction map is treated, for cells of the grid, in Lemma 16.2 below.

²⁰Lean: `IsSouzaAtom`, `canonicalSouzaAtom`, `canonicalSouzaAtom_isSouzaAtom` and the atom family `souzaAtomFamily` in `GoodGrid/BesovSpace.lean`. The paper's further examples — Hölder atoms (Section 11B) and bounded variation atoms (Section 11C) — are not formalized and are omitted.

In Part II we suppose that $s > 0$, $p \in [1, \infty)$ and $q \in [1, \infty]$.

11. BESOV SPACES IN A MEASURE SPACE WITH A GOOD GRID

We study the Besov-ish spaces $\mathcal{B}_{p,q}^s = \mathcal{B}_{p,q}^s(\mathcal{P}, \mathcal{A}_{s,p}^{sz})$ associated with a measure space with a good grid $(I, \mathcal{P}, m)$. If $p \in [1, \infty)$, $q \in [1, \infty]$ and $0 < s < \frac{1}{p}$ we say that $\mathcal{B}_{p,q}^s$ is a **Besov space**. The general theory of Part I specializes as follows.

Theorem 11.1. *Let $(I, \mathcal{P}, m)$ be a measure space with a good grid, $s > 0$, $p \in [1, \infty)$, $q \in [1, \infty]$. Then*

- A. $(\mathcal{B}_{p,q}^s, |\cdot|_{\mathcal{B}_{p,q}^s})$ is a complex Banach space.²¹
- B. Closed balls of $\mathcal{B}_{p,q}^s$ are compact subsets of L^p , and also of L^1 .²²
- C. If $0 < s < 1/p$ then $\mathcal{B}_{p,q}^s$ is dense in L^p , and dense in L^1 .²³

```

theorem souzaBesovSpace_costNorm_completeSpace
  (G : GoodGridSpace (α := α))
  (s : ℝ) (p q : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)] :
  @CompleteSpace
    (SouzaBesovSpace G s p q hs hp hp_top)
    (WeakGridSpace.BesovishSpace.costNormedAddCommGroup
      (A := souzaAtomFamily G s p hs hp hp_top) (q := q)
      hp_top (souza_hCco G s p q hs hp
        hp_top)).toMetricSpace.toPseudoMetricSpace.toUniformSpace := ...

theorem souza_closedCostBallInLp_isCompact
  (G : GoodGridSpace (α := α))
  (s : ℝ) (p q : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)]
  {C : ℝ} (hC : 0 ≤ C) :
  IsCompact
    {f : MeasureTheory.Lp C p G.toWeakGridSpace.measure |
      ∃ hf : f ∈ SouzaBesovSpace G s p q hs hp hp_top,
        WeakGridSpace.BesovishSpace.Norm_Costpq
          (souzaAtomFamily G s p hs hp hp_top) q ⟨f, hf⟩ ≤ C} := ...

theorem souzaBesovSpace_dense
  (G : GoodGridSpace (α := α))
  (s : ℝ) (p q : ℝ ≥ 0 ∞)

```

²¹Lean: `souzaBesovSpace_costNorm_completeSpace` in `GoodGrid/BesovSpace.lean`.

²²Lean: `souza_closedCostBallInLp_isCompact` and `souza_closedCostBallInL1_isCompact` in `GoodGrid/BesovSpace.lean`. This is the Souza-atom instance of Corollary 5.4 together with the embedding of Proposition 5.1.

²³Lean: `souzaBesovSpace_dense` and `souzaBesovSpace_dense_inL1` in `GoodGrid/BesovSpace.lean`. The density statements do not appear explicitly as theorems in the paper; they are recorded here because they are formalized.

```

(hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
[Fact (1 ≤ p)] [Fact (1 ≤ q)] :
Dense (SouzaBesovSpace G s p q hs hp hp_top :
  Set (MeasureTheory.Lp ℂ p G.toWeakGridSpace.measure)) := ...

```

12. POSITIVE CONE

We say that f is $\mathcal{B}_{p,q}^s$ -positive if there is a $\mathcal{B}_{p,q}^s$ -representation

$$f = \sum_k \sum_{P \in \mathcal{P}^k} c_P a_P$$

where $c_P \geq 0$ and a_P is the canonical (s, p) -Souza's atom supported on P . The set of all $\mathcal{B}_{p,q}^s$ -positive functions is a convex cone in $\mathcal{B}_{p,q}^s$, denoted $\mathcal{B}_{p,q}^{s+}$, with the “norm”

$$\|f\|_{\mathcal{B}_{p,q}^{s+}} = \inf \left(\sum_k \left(\sum_{P \in \mathcal{P}^k} c_P^p \right)^{q/p} \right)^{1/q},$$

the infimum running over all positive representations of f .²⁴

```

def SouzaPositiveRepresentation
  (G : GoodGridSpace (α := α)) (s : ℝ) (p : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)]
  {f : Lp ℂ p G.toWeakGridSpace.measure}
  (R : WeakGridSpace.LpGridRepresentation
    (souzaAtomFamily G s p hs hp hp_top) f) : Prop :=
  ∀ k, SouzaPositiveLevelBlock G s p hs hp hp_top (R.block k)

def SouzaConePositiveRepresentation
  (G : GoodGridSpace (α := α)) (s : ℝ) (p : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)]
  {f : Lp ℂ p G.toWeakGridSpace.measure}
  (R : WeakGridSpace.LpGridRepresentation
    (souzaAtomFamily G s p hs hp hp_top) f) : Prop :=
  ∀ k, SouzaConePositiveLevelBlock G s p hs hp hp_top (R.block k)

```

Proposition 12.1. *Let $0 < s < 1/p$.*

A. *If $f \in \mathcal{B}_{p,q}^s$ is (a.e.) real valued then one can find $f_+, f_- \in \mathcal{B}_{p,q}^{s+}$ such that*

$$f = f_+ - f_-,$$

²⁴Lean: `SouzaPositiveRepresentation` (canonical atoms, nonnegative coefficients) and the cone `souzaPositiveCone` in `GoodGrid/PositiveCone.lean`. The formalization also uses the wider notion of a *cone-positive* representation (`SouzaConePositiveRepresentation`): real coefficients ≥ 0 and Souza's atoms which are a.e. nonnegative real-valued, not necessarily canonical.

with the positive costs of f_+ and f_- controlled in terms of $|f|_{\mathcal{B}_{p,q}^s}$; an analogous four-term decomposition holds for complex valued f .²⁵

B. The positive cone $\mathcal{B}_{p,q}^{s+}$ is dense in the cone of nonnegative functions of L^p .²⁶

```

theorem exists_souzaPositive_decomposition_of_aeRealValued
  (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)] :
  ∃ C : ℝ,
    0 ≤ C ∧
    ∀ f : WeakGridSpace.BesovishSpace
      (souzaAtomFamily G s p hs hp hp_top) q,
      SouzaBesovAEEqRealValued G s p q hs hp hp_top f →
      ∃ u v : WeakGridSpace.BesovishSpace
        (souzaAtomFamily G s p hs hp hp_top) q,
        SouzaPositiveElement G s p q hs hp hp_top u ∧
        SouzaPositiveElement G s p q hs hp hp_top v ∧
        f = u - v ∧
        souzaPositiveNorm G s p q hs hp hp_top u ≤
          ENNReal.ofReal C *
            ENNReal.ofReal (WeakGridSpace.BesovishSpace.Norm_Costpq
              (souzaAtomFamily G s p hs hp hp_top) q f) ∧
        souzaPositiveNorm G s p q hs hp hp_top v ≤
          ENNReal.ofReal C *
            ENNReal.ofReal (WeakGridSpace.BesovishSpace.Norm_Costpq
              (souzaAtomFamily G s p hs hp hp_top) q f) := ...

theorem souzaPositiveCone_dense_in_LpNonnegativeCone
  (G : GoodGridSpace (α := α)) (s : ℝ) (β q : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hβ : 1 ≤ β) (hβ_top : β ≠ ∞)
  [Fact (1 ≤ β)] [Fact (1 ≤ q)] :
  LpNonnegativeCone G β ⊆
    closure (SouzaPositiveConeInLbeta G s β q hs hβ hβ_top) := ...

```

Proposition 12.2 (Support of positive functions). *If $f \in \mathcal{B}_{p,q}^{s+}$ then its support*

$$\text{supp } f = \{x \in I : f(x) \neq 0\}$$

*is, up to a set of zero measure, a countable union of elements of $\mathcal{P}$. Moreover, in any positive representation of f , the coefficient of a cell Q vanishes unless Q is contained (mod m) in this union.*²⁷

²⁵Lean: `exists_souzaPositive_decomposition_of_aeRealValued` and `exists_souzaPositive_decomposition_of_aeComplexValued` in `GoodGrid/PositiveCone.lean`. In the paper this appears as an unnumbered observation in Section 13; the complex case is not stated in the paper.

²⁶Lean: `souzaPositiveCone_dense_in_LpNonnegativeCone` in `GoodGrid/PositiveCone.lean`. Not stated in the paper.

²⁷Lean: `support_ae_countable_iUnion_goodGridCells_of_souzaPositiveFunction` and the support theorem `souzaPositiveRepresentation_coeff_eq_zero_of_not_subset_cell`

```

theorem support_ae_countable_iUnion_goodGridCells_of_souzaPositiveFunction
  (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)]
  {f : α → ℂ}
  (hf : SouzaPositiveFunction G s p q hs hp hp_top f) :
  IsAECountableUnionOfGoodGridCells G {x | f x ≠ 0} := ...

theorem souzaPositiveRepresentation_coeff_eq_zero_of_not_subset_cell
  (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)]
  (Qc : GoodGridCell G)
  {g : Lp ℂ p G.toWeakGridSpace.measure}
  (R : WeakGridSpace.LpGridRepresentation (souzaAtomFamily G s p hs hp
    hp_top) g)
  (hR : SouzaPositiveRepresentation G s p hs hp hp_top R)
  (hsupp : ∀m x ∂ G.toWeakGridSpace.measure, x ∉ Qc.cell → (g : α → ℂ)
    x = 0)
  {n : ℕ} (P : WeakGridSpace.LevelCell G.toWeakGridSpace n)
  (hP : ¬ P.1 ⊆ Qc.cell) :
  (R.block n).coeff P = 0 := ...

```

13. UNBALANCED HAAR WAVELETS

Let $\mathcal{P} = (\mathcal{P}^k)_k$ be a good grid. Following Girardi and Sweldens, one constructs, from the children structure of the grid (Type I tree with the logarithmic subtrees construction), a family of **unbalanced Haar functions** $\{\phi_S\}_{S \in \hat{\mathcal{H}}}$, where $\hat{\mathcal{H}} = \{I\} \cup \mathcal{H}$, $\phi_I = 1_I/|I|^{1/2}$, each ϕ_S with $S \in \mathcal{H}_Q$ is supported on Q , has zero mean, L^2 -norm one, and satisfies

$$\frac{C_5(\lambda_1, \lambda_2)}{|Q|^{1/2}} \leq |\phi_S(x)| \leq \frac{C_6(\lambda_1, \lambda_2)}{|Q|^{1/2}}$$

on its support. The family $\{\phi_S\}_{S \in \hat{\mathcal{H}}}$ is an unconditional basis of L^β for every $1 < \beta < \infty$.²⁸

```

def L2normalizedHaar (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
  (F : UnbalancedHaarWavelet.FullHaarSystem (G := GridOf G))

```

in `GoodGrid/PositiveCone.lean` (paper Proposition 13.1; the second sentence is a sharper statement obtained in the formalization).

²⁸The construction and the unconditional basis property are provided by the external Lean repositories `UnbalancedHaarWavelet`, `UnconditionalSchauderBasis`, `LaminarFamiliesMaximalBinaryTrees` and `Burkholder` (SmâniaD), on which this repository depends. The local API (L^2 -normalized Haar functions `L2normalizedHaar` and Haar coefficients `Coeff`) is in `GoodGrid/AlternativeRepresentationsAndNorms/HaarRepresentationNorm.lean`.

```

(i : F.Index) (x : α) : ℂ :=
((L2NormalizationFactor G F i : ℝ) : ℂ) *
(UnbalancedHaarWavelet.FullHaarSystem.function (GridOf G) F i x : ℂ)

def Coeff (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
(F : UnbalancedHaarWavelet.FullHaarSystem (G := GridOf G))
(f : α → ℂ) (_hf : Integrable f G.grid.μ) (i : F.Index) : ℂ :=
∫ x, f x * L2normalizedHaar G F i x ∂G.grid.μ

```

14. ALTERNATIVE CHARACTERIZATIONS I: MESSING WITH NORMS

For $f \in L^\beta$, $\beta > 1$, write $d_S^f = \int f \phi_S dm$ for the Haar coefficients of f , and define

$$N_{haar}(f) = |I|^{1/p-s-1/2} |d_I^f| + \left(\sum_k \left(\sum_{Q \in \mathcal{P}^k} |Q|^{1-sp-\frac{p}{2}} \sum_{S \in \mathcal{H}_Q} |d_S^f|^p \right)^{q/p} \right)^{1/q}.$$

From the Haar coefficients one builds, exactly as in the paper, the **standard atomic representation**

$$f = \sum_i \sum_{Q \in \mathcal{P}^i} k_Q^f a_Q$$

(a_Q canonical Souza's atoms) and its norm

$$N_{st}(f) = |k_I^f| + \left(\sum_{k \geq 1} \left(\sum_{Q \in \mathcal{P}^k} |k_Q^f|^p \right)^{q/p} \right)^{1/q}.$$

Finally, with

$$osc_p(f, Q) = \inf_{c \in \mathbb{C}} \left(\int_Q |f - c|^p dm \right)^{1/p},$$

define

$$N_{osc}(f) = |I|^{-s} |f|_p + \left(\sum_k \left(\sum_{Q \in \mathcal{P}^k} |Q|^{-sp} osc_p(f, Q)^p \right)^{q/p} \right)^{1/q}.$$

Theorem 14.1 (Equivalence of norms). *Suppose $0 < s < 1/p$, $p \in [1, \infty)$ and $q \in [1, \infty]$. Each one of the quantities $|f|_{\mathcal{B}_{p,q}^s}$, $N_{st}(f)$, $N_{haar}(f)$, $N_{osc}(f)$ is finite if and only if $f \in \mathcal{B}_{p,q}^s$, and these norms are equivalent on $\mathcal{B}_{p,q}^s$. More precisely there are constants, depending only on the grid geometry and on s, p, q , such that*

$$\begin{aligned} N_{st}(f) &\leq C |f|_{\mathcal{B}_{p,q}^s}, & N_{st}(f) &\leq C N_{haar}(f), \\ N_{haar}(f) &\leq C N_{osc}(f), & N_{osc}(f) &\leq C |f|_{\mathcal{B}_{p,q}^s}, \end{aligned}$$

and conversely: if $N_{st}(f) < \infty$ then $f \in \mathcal{B}_{p,q}^s$ and the standard atomic representation is a $\mathcal{B}_{p,q}^s$ -representation of f (so $|f|_{\mathcal{B}_{p,q}^s} \leq N_{st}(f)$); if $N_{haar}(f) < \infty$ and $f \in L^1$ then $f \in L^p$ and the Haar series of f converges in L^p .²⁹

```

theorem exists_standardRepresentationNorm_le_const_mul_haarL2RepresentationNorm
  (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
  (F : UnbalancedHaarWavelet.FullHaarSystem (G := HaarRepresentation.GridOf G))
  [DecidableEq F.Index]
  (s : ℝ) (hs : 0 < s)
  (p : ℝ≥0∞) [Fact (1 ≤ p)] (hp_top : p < ∞)
  (q : ℝ≥0∞) [Fact (1 ≤ q)] :
  ∃ C : ℝ≥0∞, C ≠ ∞ ∧
    ∀ (f : α → ℂ) (hfint : Integrable f G.grid.μ),
      HaarRepresentation.haarL2RepresentationNorm G F s p q f hfint ≠ ∞ →
        standardRepresentationNorm G F s hs p hp_top q f hfint ≠ ∞ ∧
          standardRepresentationNorm G F s hs p hp_top q f hfint ≤
            C * HaarRepresentation.haarL2RepresentationNorm G F s p q f hfint :=
    ...

theorem exists_haarL2RepresentationNorm_le_const_mul_meanOscillationNorm
  (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
  (F : UnbalancedHaarWavelet.FullHaarSystem (G := GridOf G))
  (s : ℝ) (p : ℝ≥0∞) [Fact (1 ≤ p)] (hp_top : p < ∞)
  (q : ℝ≥0∞) [Fact (1 ≤ q)] :
  ∃ C : ℝ≥0∞, C ≠ ∞ ∧
    ∀ (f : α → ℂ) (hf : Integrable f G.grid.μ), MemLp f p G.grid.μ →
      MeanOscillation.meanOscillationNorm G s p q f ≠ ∞ →
        haarL2RepresentationNorm G F s p q f hf ≤
          C * MeanOscillation.meanOscillationNorm G s p q f := ...

theorem exists_meanOscillationNorm_le_const_mul_souzaBesovNorm
  (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
  (s : ℝ) (hs : 0 < s)
  (p : ℝ≥0∞) [Fact (1 ≤ p)] (hp_top : p < ∞)
  (q : ℝ≥0∞) [Fact (1 ≤ q)] :
  ∃ C : ℝ≥0∞, C ≠ ∞ ∧
    ∀ (g : SouzaBesovSpace G s p q hs Fact.out (ne_of_lt hp_top))
      (f : α → ℂ),
      f =m[G.grid.μ] ((g : Lp ℂ p G.toWeakGridSpace.measure) : α → ℂ) →

```

²⁹Lean: the inequality cycle `exists_standardRepresentationNorm_le_const_mul_souzaBesovNorm`, `exists_standardRepresentationNorm_le_const_mul_haarL2RepresentationNorm`, `exists_haarL2RepresentationNorm_le_const_mul_meanOscillationNorm`, `exists_meanOscillationNorm_le_const_mul_souzaBesovNorm`, `finite_standardRepresentationNorm_implies_memBesov_and_standardRepresentation`, `finite_haarL2RepresentationNorm_implies_memLp_and_hasSum` and `exists_haarRepresentationNorms_equivalent`, in the files of `GoodGrid/AlternativeRepresentationsAndNorms/`. Compared with paper Theorem 15.1: the formalization proves the equivalences as a cycle of separate inequalities with existential constants (the explicit constants θ_1, θ_2 of the paper are not tracked), it works with the L^2 -normalized Haar coefficients, and there is no single wrap-up statement packaging all four norms at once.

```

meanOscillationNorm G s p q f ≤
  C * ENNReal.ofReal
    (WeakGridSpace.BesovishSpace.Norm_Costpq
      (souzaAtomFamily G s p hs Fact.out (ne_of_lt hp_top)) q g) := ...

theorem
  finite_standardRepresentationNorm_implies_memBesov_and_standardRepresentation
  (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
  (F : UnbalancedHaarWavelet.FullHaarSystem (G := HaarRepresentation.GridOf G))
  [DecidableEq F.Index]
  (s : ℝ) (hs : 0 < s)
  (p : ℝ ≥ 0∞) [Fact (1 ≤ p)] (hp_top : p < ∞)
  (q : ℝ ≥ 0∞) [Fact (1 ≤ q)]
  (f : α → ℂ) (hf : Integrable f G.grid.μ)
  (hN : standardRepresentationNorm G F s hs p hp_top q f hf ≠ ∞) :
  ∃ hflp : MemLp f p G.grid.μ,
    ∃ g : SouzaBesovSpace G s p q hs Fact.out (ne_of_lt hp_top),
      ∃ R :
        WeakGridSpace.LpGridRepresentation
          (souzaAtomFamily G s p hs Fact.out (ne_of_lt hp_top))
          (g : Lp ℂ p G.toWeakGridSpace.measure),
        (g : Lp ℂ p G.toWeakGridSpace.measure) = hflp.toLp f ∧
        R.block = canonicalStandardBlockSeq G F s hs p hp_top f hf ∧
        WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) R ∧
        WeakGridSpace.LpGridRepresentation.pqCostENNReal (q := q) R =
          standardRepresentationNorm G F s hs p hp_top q f hf ∧
        WeakGridSpace.LpGridRepresentation.pqCost (q := q) R ≤
          (standardRepresentationNorm G F s hs p hp_top q f hf).toReal ∧
        WeakGridSpace.BesovishSpace.Norm_Costpq
          (souzaAtomFamily G s p hs Fact.out (ne_of_lt hp_top)) q g
    ≤
      (standardRepresentationNorm G F s hs p hp_top q f hf).toReal
  := ...

```

Corollary 14.2 (Reading the standard coefficients). *For each $P \in \mathcal{P}$ the coefficient k_P^f of the standard atomic representation of f is a (finite) linear combination of the Haar coefficients d_S^f , $S \in \mathcal{H}_Q$, of f ,³⁰ and the standard representation of any $f \in \mathcal{B}_{p,q}^s$ converges to f in L^p with*

$$\left(\sum_k \left(\sum_{P \in \mathcal{P}^k} |k_P^f|^p \right)^{q/p} \right)^{1/q} \leq C \|f\|_{\mathcal{B}_{p,q}^s}.$$

```

theorem hasSum_coeff_smul_l2normalizedHaar_toLp
  (G : GoodGridSpace (α := α)) [DecidableEq (Set α)]
  (F : UnbalancedHaarWavelet.FullHaarSystem (G := GridOf G))

```

³⁰Lean: `Coeff` and `hasSum_coeff_smul_l2normalizedHaar.toLp` in `GoodGrid/AlternativeRepresentationsAndNorms/HaarRepresentationNorm.lean`, together with the theorems of the previous footnote. Paper Corollary 15.2 states moreover that $f \mapsto k_P^f$ extends to a bounded linear functional on L^1 ; this extension is not formalized.

```

[DecidableEq F.Index]
(β : ℝ≥0∞) (hβ_one : 1 < β) (hβ_top : β < ∞)
(f : α → ℂ) (hf : MemLp f β G.grid.μ) :
letI : Fact (1 ≤ β) := ⟨le_of_lt hβ_one⟩
HasSum
  (fun i : F.Index =>
    Coeff G F f
      (by
        letI : IsFiniteMeasure G.grid.μ := (GridOf G).isFinite
        exact hf.integrable (le_of_lt hβ_one))
    i •
    (l2normalizedHaar_memLp G F β i).toLp (L2normalizedHaar G F i))
  (hf.toLp f) := ...

```

15. ALTERNATIVE CHARACTERIZATIONS II: MESSING WITH ATOMS

Let $s < \beta$ and $\tilde{q} \in [1, \infty]$. A $(s, \beta, p, \tilde{q})$ -**Besov atom** on Q is a function $a \in \mathcal{B}_{p,\tilde{q}}^\beta(Q, \mathcal{P}_Q, \mathcal{A}_{\beta,p}^{sz})$ such that $a(x) = 0$ for $x \notin Q$ and

$$|a|_{\mathcal{B}_{p,\tilde{q}}^\beta(Q, \mathcal{P}_Q, \mathcal{A}_{\beta,p}^{sz})} \leq \frac{1}{C_7} |Q|^{s-\beta},$$

where $C_7 = C_7(C_1, \lambda_2, \beta, \tilde{q}, p) \geq 1$ is the normalizing constant of the paper. The family of $(s, \beta, p, \tilde{q})$ -Besov atoms supported on Q is denoted $\mathcal{A}_{s,\beta,p,\tilde{q}}^{bs}(Q)$, and $\mathcal{A}_{s,p}^{sz}(Q) \subset \mathcal{A}_{s,\beta,p,\tilde{q}}^{bs}(Q)$; Besov atoms are atoms of type $(s, p, 1)$.

Proposition 15.1 (Souza's atoms and Besov's atoms). *Let $\mathcal{P}$ be a good grid and $0 < s < 1/p$. Let $\mathcal{A}$ be a class of (s, p, u) -atoms, with $u \geq 1$, such that for some $s < \beta$, $\tilde{q} \in [1, \infty]$, and $C_8, C_9 \geq 0$ we have, for every $Q \in \mathcal{P}$,*

$$\frac{1}{C_8} \mathcal{A}_{s,p}^{sz}(Q) \subset \mathcal{A}(Q) \subset C_9 \mathcal{A}_{s,\beta,p,\tilde{q}}^{bs}(Q).$$

Then³¹

$$\mathcal{B}_{p,q}^s(\mathcal{P}, \mathcal{A}_{s,p}^{sz}) = \mathcal{B}_{p,q}^s(\mathcal{P}, \mathcal{A}) = \mathcal{B}_{p,q}^s(\mathcal{P}, \mathcal{A}_{s,\beta,p,\tilde{q}}^{bs}),$$

with

$$|f|_{\mathcal{B}_{p,q}^s(\mathcal{A})} \leq C_8 |f|_{\mathcal{B}_{p,q}^s(\mathcal{A}_{s,p}^{sz})} \quad \text{and} \quad |f|_{\mathcal{B}_{p,q}^s(\mathcal{A}_{s,\beta,p,\tilde{q}}^{sz})} \leq C |f|_{\mathcal{B}_{p,q}^s(\mathcal{A})}.$$

³¹Lean: `atoms_between_souza_atoms_and_besov_atoms` in `GoodGrid/BesovAtoms.lean`, supported by the decay claims `besovAtom_to_souza_representation_decay` and `besovAtom_to_induced_souzaS_representation_decay_claimC` (the Lean counterparts of the Claim in the paper's proof, with the geometric decay rate $\lambda_2^{(k-j)(\beta-s)}$). Paper Propositions 16.2 (Hölder atoms) and 16.3 (bounded variation atoms) are not formalized and are omitted.


```

theorem atoms_between_souza_atoms_and_besov_atoms
  (G : GoodGridSpace (α := α)) (s β : ℝ) (p u q qtilde : ℝ≥0∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ u)] [Fact (1 ≤ q)] [Fact (1 ≤ qtilde)]
  (hs : 0 < s) (hβ : 0 < β) (hβs : s < β) (hp_top : p ≠ ∞)
  (A : WeakGridSpace.AtomFamily G.toWeakGridSpace s p u)
  (C1 C2 : ℝ)
  (hSandwich :
    SouzaBesovSandwich G s β p u qtilde hs hβ (inferInstance : Fact (1 ≤ p))
    hp_top A C1 C2) :
  (WeakGridSpace.BesovishSpace (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p)
    hp_top) q =
    WeakGridSpace.BesovishSpace A q) ∧
  (WeakGridSpace.BesovishSpace A q =
    WeakGridSpace.BesovishSpace
      (besovAtomFamily G s β p qtilde hs hβ inferInstance hp_top) q) ∧
  (∀ f : WeakGridSpace.BesovishSpace
    (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p) hp_top) q,
    ∃ hfA : WeakGridSpace.MemBesovishCoeffCost A q
      (f : Lp ℂ p G.toWeakGridSpace.measure),
      WeakGridSpace.BesovishSpace.Norm_Costpq A q
        ((f : Lp ℂ p G.toWeakGridSpace.measure), hfA) :
        WeakGridSpace.BesovishSpace A q)
      ≤ C1 *
        WeakGridSpace.BesovishSpace.Norm_Costpq
          (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p) hp_top) q f) ∧
  (∀ f : WeakGridSpace.BesovishSpace A q,
    ∃ hfS : WeakGridSpace.MemBesovishCoeffCost
      (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p) hp_top) q
      (f : Lp ℂ p G.toWeakGridSpace.measure),
      WeakGridSpace.BesovishSpace.Norm_Costpq
        (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p) hp_top) q
        ((f : Lp ℂ p G.toWeakGridSpace.measure), hfS) :
        WeakGridSpace.BesovishSpace
          (souzaAtomFamily G s p hs (Fact.out : 1 ≤ p) hp_top) q)
      ≤ (C2 / (1 - G.grid.lambda2 ^ (β - s))) *
        WeakGridSpace.BesovishSpace.Norm_Costpq A q f) := ...

```

III. APPLICATIONS

In Part III we suppose that $0 < s < 1/p$, $p \in [1, \infty)$ and $q \in [1, \infty]$.

16. POINTWISE MULTIPLIERS ACTING ON $\mathcal{B}_{p,q}^s$

Let $g: I \rightarrow \mathbb{C}$ be a measurable function. We say that g is a **pointwise multiplier** acting on $\mathcal{B}_{p,q}^s$ if the transformation $G(f) = gf$ defines a bounded operator in $\mathcal{B}_{p,q}^s$. We denote the set of pointwise multipliers by $M(\mathcal{B}_{p,q}^s)$, with

the norm

$$|g|_{M(\mathcal{B}_{p,q}^s)} = \sup\{|gf|_{\mathcal{B}_{p,q}^s} : |f|_{\mathcal{B}_{p,q}^s} \leq 1\}.$$

Define

$$\mathcal{B}_{p,q, selfs}^s = \left\{ g \in \mathcal{B}_{p,q}^s : \sup_{a_Q \in \mathcal{A}_{s,p}^{sz}} |ga_Q|_{\mathcal{B}_{p,q}^s} < \infty \right\},$$

with the norm

$$|g|_{\mathcal{B}_{p,q, selfs}^s} = \sup_{a_Q \in \mathcal{A}_{s,p}^{sz}} |ga_Q|_{\mathcal{B}_{p,q}^s},$$

and, for $t \in \mathbb{N}$, the tail seminorms³²

$$|g|_{\mathcal{B}_{p,q, selfs}^{s,t}} = \sup_{\substack{a_Q \in \mathcal{A}_{s,p}^{sz} \\ Q \in \mathcal{P}^j, j \geq t}} |ga_Q|_{\mathcal{B}_{p,q}^s}.$$

```

abbrev SouzaPointwiseMultiplier
  (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)]
  (m : α → ℂ) : Prop :=
  WeakGridSpace.IsPointwiseMultiplier
  (A := souzaAtomFamily G s p hs hp hp_top) q m

abbrev SouzaPointwiseSelfsClass
  (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)]
  (m : α → ℂ) : Prop :=
  WeakGridSpace.PointwiseSelfsClass
  (A := souzaAtomFamily G s p hs hp hp_top) q m

```

Proposition 16.1. *We have that*³³

$$M(\mathcal{B}_{1,1}^s) = \mathcal{B}_{1,1, selfs}^s.$$

```

theorem souzaPointwiseMultiplier_iff_souzaPointwiseSelfsClass_one_one
  (G : GoodGridSpace (α := α)) (s : ℝ)
  (hs : 0 < s) {m : α → ℂ} :
  SouzaPointwiseMultiplier G s (1 : ℝ ≥ 0 ∞) (1 : ℝ ≥ 0 ∞)

```

³²Lean: SouzaPointwiseMultiplier, SouzaPointwiseSelfsClass and the tail class SouzaPointwiseSelfsTailClass with their seminorms, in GoodGrid/Multipliers/Definition.lean.

³³Lean: souzaPointwiseMultiplier_iff_souzaPointwiseSelfsClass_one_one (the inclusion $\mathcal{B}_{1,1, selfs}^s \subset M(\mathcal{B}_{1,1}^s)$ is proved in GoodGrid/Multipliers/Besovs11.lean; the converse, which holds for all p, q , is souzaPointwiseSelfsClass_of_souzaPointwiseMultiplier in GoodGrid/Multipliers/Besovspq.lean). This is paper Proposition 18.1.

```

hs le_rfl ENNReal.one_ne_top m ↔
SouzaPointwiseSelfsClass G s (1 : ℝ≥0∞) (1 : ℝ≥0∞)
hs le_rfl ENNReal.one_ne_top m := ...

```

Lemma 16.2 (Restriction to a cell). *Let $W \in \mathcal{P}$. The restriction map*

$$r: \mathcal{B}_{p,q}^s(I, \mathcal{P}, \mathcal{A}_{s,p}^{sz}) \rightarrow \mathcal{B}_{p,q}^s(W, \mathcal{P}_W, \mathcal{A}_{s,p}^{sz}), \quad r(f) = 1_W f,$$

is continuous: there is $C_{10} \geq 1$, independent of W , such that

A. *For every $f \in \mathcal{B}_{p,q}^s$ we have*

$$|1_W f|_{\mathcal{B}_{p,q}^s(W, \mathcal{P}_W, \mathcal{A}_{s,p}^{sz})} \leq C_{10} |f|_{\mathcal{B}_{p,q}^s}.$$

In particular $1_W \in M(\mathcal{B}_{p,q}^s)$ with $|1_W|_{M(\mathcal{B}_{p,q}^s)} \leq C_{10}$, uniformly in W .

B. *For every $\mathcal{B}_{p,q}^{s+}$ -representation $f = \sum_k \sum_Q c_Q a_Q$ one can find a positive representation*

$$1_W f = \sum_k \sum_Q d_Q a_Q$$

*on the induced grid with cost at most C_{10} times the cost of the original one, and such that $d_Q \neq 0$ implies $Q \subset \text{supp } f \pmod{m}$.*³⁴

```

theorem souzaCellIndicatorPointwiseMultiplier
  (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ≥0∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)]
  (hs_lt_inv : s < (p.toReal)-1)
  (W : GoodGridCell G) :
  SouzaPointwiseMultiplier G s p q hs hp hp_top
  (W.cell.indicator fun _ => (1 : ℂ)) := ...

theorem souzaCellIndicatorPointwiseMultiplierBound_uniform
  (G : GoodGridSpace (α := α)) (s : ℝ) (p q : ℝ≥0∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)]
  (hs_lt_inv : s < (p.toReal)-1)
  (W : GoodGridCell G) :
  SouzaPointwiseMultiplierBound G s p q hs hp hp_top
  (W.cell.indicator fun _ => (1 : ℂ))
  (2 * souzaAmbientRestrictionMultiplierConstant G s p + 1) := ...

```

³⁴Lean: `souzaCellIndicatorRestrictsToInduced`, the bounds `souzaIndicatorRestrictionBound_of_ambientRestrictionTransmutation*`, `souzaCellIndicatorPointwiseMultiplierBound_uniform` and `souzaCellIndicatorPointwiseMultiplier` in `GoodGrid/Multipliers/Besovspq.lean` (paper Lemma 18.2). The paper's strongly regular domains (Definition 18.7 and Proposition 18.8), of which the present lemma is the grid-cell case, are not formalized in general; the multiplier statement for 1_W , $W \in \mathcal{P}$, with a bound uniform in W , is the formalized special case.

Proposition 16.3 (*selfs* classes are bounded). *There is $C_{11} = C_{11}(\lambda_1, \lambda_2, C_1, s, p, q)$ such that for every $t \in \mathbb{N}$ and every g with $|g|_{\mathcal{B}_{p,q,\text{selfs}}^{s,t}} < \infty$,*

$$|g(x)| \leq C_{11} |g|_{\mathcal{B}_{p,q,\text{selfs}}^{s,t}} \quad \text{for } m\text{-a.e. } x.$$

*In particular $\mathcal{B}_{p,q,\text{selfs}}^s \subset L^\infty$ and this inclusion is continuous.*³⁵

```

theorem souzaPointwiseSelfsTailNorm_norm_ae_le
  (G : GoodGridSpace (α := α))
  (s : ℝ) (p q : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)]
  (hs_lt_inv : s < (p.toReal)-1)
  {t : ℕ} {m : α → ℂ}
  (hm : SouzaPointwiseSelfsTailClass G s p q hs hp hp_top t m) :
  ∀m x ∂G.grid.μ,
    ||m x|| ≤
      souzaBesovLpLocalEmbeddingConstant G s p q *
      (2 * souzaAmbientRestrictionMultiplierConstant G s p + 1) *
      souzaPointwiseSelfsTailNorm G s p q hs hp hp_top t m := ...

```

16.1. Non-Archimedean behaviour in $\mathcal{B}_{p,\tilde{q},\text{selfs}}^\beta$. If $g_i \in M(\mathcal{B}_{p,q}^s)$, the naive estimate is

$$\left| \left(\sum_i g_i \right) f \right|_{\mathcal{B}_{p,q}^s} \leq \left(\sum_i |g_i|_{M(\mathcal{B}_{p,q}^s)} \right) |f|_{\mathcal{B}_{p,q}^s}.$$

Remarkably, when the supports of the g_i are separated by the grid, one gets a far better estimate.

Theorem 16.4 (Non-Archimedean property, finite version). *Let $\beta > s$ and $\tilde{q} \in [1, \infty]$. There is $C_{12} = C_{12}(\lambda_1, \lambda_2, C_1, s, \beta, p, q, \tilde{q})$ with the following property. Let Λ be finite, g_i with $|g_i|_{\mathcal{B}_{p,\tilde{q},\text{selfs}}^{\beta,t_i}} < \infty$ for $i \in \Lambda$ and $t_i \in \mathbb{N}$. Consider a function f with a $\mathcal{B}_{p,q}^s$ -representation*

$$f = \sum_k \sum_{Q \in \mathcal{P}^k} c_Q a_Q$$

by canonical Souza's atoms satisfying

³⁵Lean: `souzaPointwiseSelfsTailBound_norm_ae_le`, `souzaPointwiseSelfsTailNorm_norm_ae_le` and `souzaPointwiseSelfsTailClass_norm_ae_bounded` in `GoodGrid/Multipliers/MultipliersareBounded.lean`. This sharpens paper Proposition 18.3 in two respects: the estimate is proved for the tail seminorms $|\cdot|_{\mathcal{B}_{p,q,\text{selfs}}^{s,t}}$ with a constant *independent of t* (this uniformity is what the non-Archimedean arguments below require), and the formalization also provides a *canonical* variant (`SouzaPointwiseCanonicalSelfsTailBound`) in which g is only tested against canonical Souza's atoms.

A. We have

$$\sup_{\substack{Q \in \mathcal{P} \\ c_Q \neq 0}} \sum_{i: Q \cap \text{supp } g_i \neq \emptyset} |g_i|_{\mathcal{B}_{p,q,\text{self}}^{\beta,t_i}} \leq N.$$

B. If $Q \in \mathcal{P}^k$ satisfies $c_Q \neq 0$ and $Q \cap \text{supp } g_i \neq \emptyset$ then $k \geq t_i$.

Then $(\sum_{i \in \Lambda} g_i)f \in \mathcal{B}_{p,q}^s$ and we can find a $\mathcal{B}_{p,q}^s$ -representation³⁶

$$(7) \quad \left(\sum_{i \in \Lambda} g_i \right) f = \sum_k \sum_{Q \in \mathcal{P}^k} d_Q a_Q$$

by canonical Souza's atoms such that

$$(8) \quad \left(\sum_k \left(\sum_{Q \in \mathcal{P}^k} |d_Q|^p \right)^{q/p} \right)^{1/q} \leq C_{12} N \left(\sum_k \left(\sum_{Q \in \mathcal{P}^k} |c_Q|^p \right)^{q/p} \right)^{1/q}.$$

```

theorem souzaNonArchimedeanPropertyLambdaFinite
  (G : GoodGridSpace (α := α))
  (s β : ℝ) (p q qtilde : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hβ : 0 < β) (hβs : s < β)
  (hβ_lt_inv : β < (p.toReal)-1)
  (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)] [Fact (1 ≤ qtilde)] :
  ∃ Cgen : ℝ,
    0 ≤ Cgen ∧
    ∀ (Λ : Finset ℕ) (t : ℕ → ℕ) (g : ℕ → α → ℂ) (N : ℝ)
      (f : α → ℂ)
      (x : WeakGridSpace.BesovishSpace
        (souzaAtomFamily G s p hs hp hp_top) q)
      (R : WeakGridSpace.LpGridRepresentation
        (souzaAtomFamily G s p hs hp hp_top)
        (x : Lp ℂ p G.toWeakGridSpace.measure)),
    0 ≤ N →
    WeakGridSpace.RepresentsFunction
      (G := G.toWeakGridSpace) (p := p) f
      (x : Lp ℂ p G.toWeakGridSpace.measure) →
    WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) R →
    (∀ i ∈ Λ,
      ∃ C : ℝ,
        SouzaPointwiseSelfsTailBound
          G β p qtilde hβ hp hp_top (t i) (g i) C) →
    (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
      (R.block k).coeff Q ≠ 0 →
        nonArchimedeanRelevantTailSelfsSum
          G β p qtilde hβ hp hp_top ∧ t g Q ≤ N) →
    (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k) i,

```

³⁶Lean: `souzaNonArchimedeanPropertyLambdaFinite` in `GoodGrid/Multipliers/NonArchimedeanProperty.lean` (paper Proposition 18.4 for finite Λ). The representations of f and of the product are by *canonical* Souza's atoms (`SouzaCanonicalRepresentation`), which is no loss of generality for Souza's atoms and is the precise form formalized.

```

i ∈ Λ →
(R.block k).coeff Q ≠ 0 →
goodGridLevelCellMeetsSupport G Q (g i) →
t i ≤ k) →
∃ y : WeakGridSpace.BesovishSpace
(souzaAtomFamily G s p hs hp hp_top) q,
∃ S : WeakGridSpace.LpGridRepresentation
(souzaAtomFamily G s p hs hp hp_top)
(y : Lp ℂ p G.toWeakGridSpace.measure),
WeakGridSpace.RepresentsFunction
(G := G.toWeakGridSpace) (p := p)
(fun z => (∑ i ∈ Λ, g i z) * f z)
(y : Lp ℂ p G.toWeakGridSpace.measure) ∧
WeakGridSpace.LpGridRepresentation.pqCost (q := q) S ≤
Cgen * N *
WeakGridSpace.LpGridRepresentation.pqCost (q := q) R := ...

```

Theorem 16.5 (Non-Archimedean property, infinite version). *In the setting of Theorem 16.4 let $\Lambda \subset \mathbb{N}$ be infinite, and replace hypothesis A. by the summability condition*

$$\sum_{i \in \Lambda} T_i \leq N, \quad T_i := \sup_{\substack{Q \in \mathcal{P}, c_Q \neq 0 \\ Q \cap \text{supp } g_i \neq \emptyset}} |g_i|_{B_{p, \tilde{q}, \text{selfs}}^{\beta, t_i}},$$

understood as a series with values in $[0, \infty]$. Then, for m-a.e. x in $\{f \neq 0\}$, the series $\sum_{i \in \Lambda} g_i(x)$ converges absolutely, with

$$\sum_{i \in \Lambda} |g_i(x)| \leq C_{12} N;$$

the function $h = (\sum_{i \in \Lambda} g_i)f$ (defined a.e. as this pointwise limit times f) belongs to L^p and to $\mathcal{B}_{p, q}^s$, and it has a canonical representation (7) satisfying (8).³⁷

```

theorem souzaNonArchimedeanProperty
(G : GoodGridSpace (α := α))
(s β : ℝ) (p q qtilde : ℝ ≥ 0 ∞)
(hs : 0 < s) (hβ : 0 < β) (hβs : s < β)
(hβ_lt_inv : β < (p.toReal)-1)
(hp : 1 ≤ p) (hp_top : p ≠ ∞)
[Fact (1 ≤ p)] [Fact (1 ≤ q)] [Fact (1 ≤ qtilde)] :
∃ Cgen : ℝ,
0 ≤ Cgen ∧
1 ≤ Cgen ∧

```

³⁷Lean: `souzaNonArchimedeanProperty` in `GoodGrid/Multipliers/NonArchimedeanProperty.lean`.

Paper Proposition 18.4 treats finite and infinite Λ together, reducing to the finite case; the formalization keeps them separate because the infinite case has genuinely more content (pointwise absolute convergence, membership of the limit in L^p , and identification of the limit, obtained through the compactness of representations with uniform cost, `exists_strongly_convergent_subseq_of_uniform_pqCost`).

```

∀ (Λ : Set ℕ) (t : ℕ → ℕ) (g : ℕ → α → ℂ) (N : ℝ)
(f : α → ℂ)
(x : WeakGridSpace.BesovishSpace
  (souzaAtomFamily G s p hs hp hp_top) q)
(R : WeakGridSpace.LpGridRepresentation
  (souzaAtomFamily G s p hs hp hp_top)
  (x : Lp ℂ p G.toWeakGridSpace.measure)),
0 ≤ N →
WeakGridSpace.RepresentsFunction
  (G := G.toWeakGridSpace) (p := p) f
  (x : Lp ℂ p G.toWeakGridSpace.measure) →
WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) R →
(∀ i ∈ Λ,
  ∃ C : ℝ,
    SouzaPointwiseSelfsTailBound
      G β p qtilde hβ hp hp_top (t i) (g i) C) →
(∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
  (R.block k).coeff Q ≠ 0 →
    ∃ T : ℝ,
      HasSum
        (fun i : {i // i ∈ Λ} =>
          nonArchimedeanRelevantTailSelfsInfiniteTerm
            G β p qtilde hβ hp hp_top Λ t g Q i)
        T ∧
        T ≤ N) →
  (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k) i,
    i ∈ Λ →
      (R.block k).coeff Q ≠ 0 →
        goodGridLevelCellMeetsSupport G Q (g i) →
          t i ≤ k) →
  ∃ h : α → ℂ,
    ∃ absSum : α → ℝ,
      (∀m z ∂G.toWeakGridSpace.measure,
        f z ≠ 0 →
          HasSum
            (fun i : {i // i ∈ Λ} => ||g i.1 z||)
            (absSum z) ∧
            absSum z ≤ Cgen * N) ∧
      (∀m z ∂G.toWeakGridSpace.measure,
        HasSum
          (fun i : {i // i ∈ Λ} => g i.1 z * f z)
          (h z)) ∧
      (∀m z ∂G.toWeakGridSpace.measure,
        ||h z|| ≤ Cgen * N * ||f z||) ∧
      (∃ hmem : MemLp h p G.toWeakGridSpace.measure,
        ||MemLp.toLp h hmem|| ≤
          Cgen * N * ||(x : Lp ℂ p G.toWeakGridSpace.measure)||) ∧
  ∃ y : WeakGridSpace.BesovishSpace
    (souzaAtomFamily G s p hs hp hp_top) q,
  ∃ S : WeakGridSpace.LpGridRepresentation
    (souzaAtomFamily G s p hs hp hp_top)
    (y : Lp ℂ p G.toWeakGridSpace.measure),
    WeakGridSpace.RepresentsFunction
      (G := G.toWeakGridSpace) (p := p) h
      (y : Lp ℂ p G.toWeakGridSpace.measure) ∧

```

```

WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) S ∧
WeakGridSpace.LpGridRepresentation.pqCost (q := q) S ≤
Cgen * N *
WeakGridSpace.LpGridRepresentation.pqCost (q := q) R := ...

```

Theorem 16.6 (Non-Archimedean property, positive version). *In the setting of Theorems 16.4 and 16.5 (finite or infinite Λ , with the respective hypotheses, the tail seminorms being replaced by their positive-cone counterparts*

$$|g|_{\mathcal{B}_{p,\tilde{q},self}^{\beta+,t}} = \sup_{\substack{a_Q \in \mathcal{A}_{\beta,p}^{zz} \\ Q \in \mathcal{P}^j, j \geq t}} |ga_Q|_{\mathcal{B}_{p,\tilde{q}}^{\beta+}}$$

for the functions g_i), assume that each g_i is $\mathcal{B}_{p,\tilde{q}}^{\beta,t_i}$ -positive. Let R be the given canonical representation $(c_Q)_Q$ of f . Then the representation $S = (d_Q)_Q$ of $(\sum_i g_i)f$ produced by Theorems 16.4/16.5 can be chosen so that, in addition:

- i. (positivity) if R is a positive representation ($c_Q \geq 0$ for all Q), then S is a cone-positive representation: all d_Q are nonnegative reals and the corresponding Souza's atoms are a.e. nonnegative;
- ii. (support) unconditionally — without assuming $c_Q \geq 0$ — every cell Q active in S ($d_Q \neq 0$) satisfies: there exists $i \in \Lambda$ with $g_i \neq 0$ a.e. on Q .³⁸

```

theorem souzaNonArchimedeanPropertyPositiveCone
  (G : GoodGridSpace (α := α))
  (s β : ℝ) (p q qtilde : ℝ ≥ 0 ∞)
  (hs : 0 < s) (hβ : 0 < β) (hβs : s < β)
  (hβ_lt_inv : β < (p.toReal)-1)
  (hp : 1 ≤ p) (hp_top : p ≠ ∞)
  [Fact (1 ≤ p)] [Fact (1 ≤ q)] [Fact (1 ≤ qtilde)] :
  ∃ Cgen : ℝ,
    0 ≤ Cgen ∧
    ∀ (Λ : Finset N) (t : N → N) (g : N → α → ℂ) (N : ℝ)
      (f : α → ℂ)

```

³⁸Lean: `souzaNonArchimedeanPropertyPositiveCone` (finite Λ) and `souzaNonArchimedeanPropertyPositiveConeInfinite` ($\Lambda \subset \mathbb{N}$ infinite, with condition Λ as an $[0, \infty]$ -valued series bound) in `GoodGrid/Multipliers/NonArchimedeanPropertyPositiveStandalone.lean`, with cores in `NonArchimedeanProperty.lean` and `GoodGrid/PositiveCone.lean`; the auxiliary product representation is `exists_nonArchimedeanProductRepresentation_positive`. Compared with paper Remark 18.5 the statement was made precise as follows: (a) the hypothesis on the representation R of f is only that it is canonical (arbitrary complex coefficients); (b) the two conclusions are separated according to their true strengths — the support conclusion ii. holds unconditionally, and is weakened to its a.e. form ($g_i \neq 0$ a.e. on Q rather than $Q \subset \text{supp } g_i$); (c) the positivity conclusion i. produces a cone-positive representation (nonnegative real coefficients, a.e. nonnegative Souza's atoms), not necessarily one by canonical atoms. Axioms of all four non-Archimedean theorems checked: `propext`, `Classical.choice`, `Quot.sound`.


```

(x : WeakGridSpace.BesovishSpace
 (souzaAtomFamily G s p hs hp hp_top) q)
(R : WeakGridSpace.LpGridRepresentation
 (souzaAtomFamily G s p hs hp hp_top)
 (x : Lp  $\mathbb{C}$  p G.toWeakGridSpace.measure)),
0 ≤ N →
WeakGridSpace.RepresentsFunction
 (G := G.toWeakGridSpace) (p := p) f
 (x : Lp  $\mathbb{C}$  p G.toWeakGridSpace.measure) →
WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) R →
SouzaCanonicalRepresentation G s p hs hp hp_top R →
(∀ i ∈ Λ,
 SouzaPositiveFunction G β p qtilde hβ hp hp_top (g i)) →
(∀ i ∈ Λ,
 ∃ C :  $\mathbb{R}_{\geq 0\infty}$ ,
 SouzaPositivePointwiseSelfsTailBound
 G β p qtilde hβ hp hp_top (t i) (g i) C) →
(∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
 (R.block k).coeff Q ≠ 0 →
 nonArchimedeanRelevantPositiveTailSelfsSum
 G β p qtilde hβ hp hp_top ∧ t g Q ≤ ENNReal.ofReal N) →
(∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k) i,
 i ∈ Λ →
 (R.block k).coeff Q ≠ 0 →
 goodGridLevelCellMeetsSupport G Q (g i) →
 t i ≤ k) →
∃ y : WeakGridSpace.BesovishSpace
 (souzaAtomFamily G s p hs hp hp_top) q,
∃ S : WeakGridSpace.LpGridRepresentation
 (souzaAtomFamily G s p hs hp hp_top)
 (y : Lp  $\mathbb{C}$  p G.toWeakGridSpace.measure),
WeakGridSpace.RepresentsFunction
 (G := G.toWeakGridSpace) (p := p)
 (fun z => (∑ i ∈ Λ, g i z) * f z)
 (y : Lp  $\mathbb{C}$  p G.toWeakGridSpace.measure) ∧
WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) S ∧
WeakGridSpace.LpGridRepresentation.pqCost (q := q) S ≤
 Cgen * N *
 WeakGridSpace.LpGridRepresentation.pqCost (q := q) R ∧
-- [ii] support: holds with only canonical atoms on 'R' (no sign on
'c_Q').
(∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
 (S.block k).coeff Q ≠ 0 →
 ∃ i ∈ Λ, ∀m z ∂(G.toWeakGridSpace.measure.restrict Q.1), g i z
 ≠ 0) ∧
-- [i] positivity: only when 'R' is fully positive (canonical + 'c_Q
≥ 0').
(SouzaPositiveRepresentation G s p hs hp hp_top R →
 SouzaConePositiveRepresentation G s p hs hp hp_top S) := ...

theorem souzaNonArchimedeanPropertyPositiveConeInfinite
 (G : GoodGridSpace (α := α))
 (s β : ℝ) (p q qtilde :  $\mathbb{R}_{\geq 0\infty}$ )
 (hs : 0 < s) (hβ : 0 < β) (hβs : s < β)
 (hβ_lt_inv : β < (p.toReal)-1)

```

```

(hp : 1 ≤ p) (hp_top : p ≠ ∞)
[Fact (1 ≤ p)] [Fact (1 ≤ q)] [Fact (1 ≤ qtilde)] :
∃ Cgen : ℝ,
  0 ≤ Cgen ∧
  1 ≤ Cgen ∧
  ∀ (Λ : Set ℕ) (t : ℕ → ℕ) (g : ℕ → α → ℂ) (N : ℝ)
    (f : α → ℂ)
    (x : WeakGridSpace.BesovishSpace
      (souzaAtomFamily G s p hs hp hp_top) q)
    (R : WeakGridSpace.LpGridRepresentation
      (souzaAtomFamily G s p hs hp hp_top)
      (x : Lp ℂ p G.toWeakGridSpace.measure)),
    0 ≤ N →
    WeakGridSpace.RepresentsFunction
      (G := G.toWeakGridSpace) (p := p) f
      (x : Lp ℂ p G.toWeakGridSpace.measure) →
    WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) R →
    SouzaCanonicalRepresentation G s p hs hp hp_top R →
    (∀ i ∈ Λ,
      SouzaPositiveFunction G β p qtilde hβ hp hp_top (g i)) →
    (∀ i ∈ Λ,
      ∃ C : ℝ ≥ 0∞,
        SouzaPositivePointwiseSelfsTailBound
          G β p qtilde hβ hp hp_top (t i) (g i) C) →
    (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
      (R.block k).coeff Q ≠ 0 →
        (∑' i : {i // i ∈ Λ},
          nonArchimedeanRelevantPositiveTailSelfsInfiniteTerm
            G β p qtilde hβ hp hp_top Λ t g Q i) ≤ ENNReal.ofReal N) →
    (∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k) i,
      i ∈ Λ →
        (R.block k).coeff Q ≠ 0 →
          goodGridLevelCellMeetsSupport G Q (g i) →
            t i ≤ k) →
    ∃ h : α → ℂ,
    ∃ absSum : α → ℝ,
      (∀m z ∂G.toWeakGridSpace.measure,
        f z ≠ 0 →
          HasSum
            (fun i : {i // i ∈ Λ} => ||g i.1 z||)
            (absSum z) ∧
            absSum z ≤ Cgen * N) ∧
      (∀m z ∂G.toWeakGridSpace.measure,
        HasSum
          (fun i : {i // i ∈ Λ} => g i.1 z * f z)
          (h z)) ∧
      (∀m z ∂G.toWeakGridSpace.measure,
        ||h z|| ≤ Cgen * N * ||f z||) ∧
      (∃ hmem : MemLp h p G.toWeakGridSpace.measure,
        ||MemLp.toLp h hmem|| ≤
          Cgen * N * ||(x : Lp ℂ p G.toWeakGridSpace.measure)||) ∧
    ∃ y : WeakGridSpace.BesovishSpace
      (souzaAtomFamily G s p hs hp hp_top) q,
    ∃ S : WeakGridSpace.LpGridRepresentation
      (souzaAtomFamily G s p hs hp hp_top)

```

```

(y : Lp C p G.toWeakGridSpace.measure),
WeakGridSpace.RepresentsFunction
(G := G.toWeakGridSpace) (p := p) h
(y : Lp C p G.toWeakGridSpace.measure) ∧
WeakGridSpace.LpGridRepresentation.FinitePQCost (q := q) S ∧
WeakGridSpace.LpGridRepresentation.pqCost (q := q) S ≤
Cgen * N *
WeakGridSpace.LpGridRepresentation.pqCost (q := q) R ∧
(∀ k (Q : WeakGridSpace.LevelCell G.toWeakGridSpace k),
(S.block k).coeff Q ≠ 0 →
∃ i ∈ Λ,
∀m z ∂(G.toWeakGridSpace.measure.restrict Q.1),
g i z ≠ 0) ∧
(SouzaPositiveRepresentation G s p hs hp hp_top R →
SouzaConePositiveRepresentation G s p hs hp hp_top S) := ...

```

Remark 16.7. Paper Corollary 18.6 ($\mathcal{B}_{p,\tilde{q},selfs}^\beta \subset M(\mathcal{B}_{p,q}^s)$ continuously, for $\beta > s$) follows from the formalized Theorem 16.4 exactly as in the paper, but it is not itself stated in the repository; it is therefore omitted from this document.

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